

Uncertainty visualization

CS424: Visualization & Visual Analytics
Fabio Miranda

Slides based on Carolina Veiga's uncertainty class



urbantk.org



evl

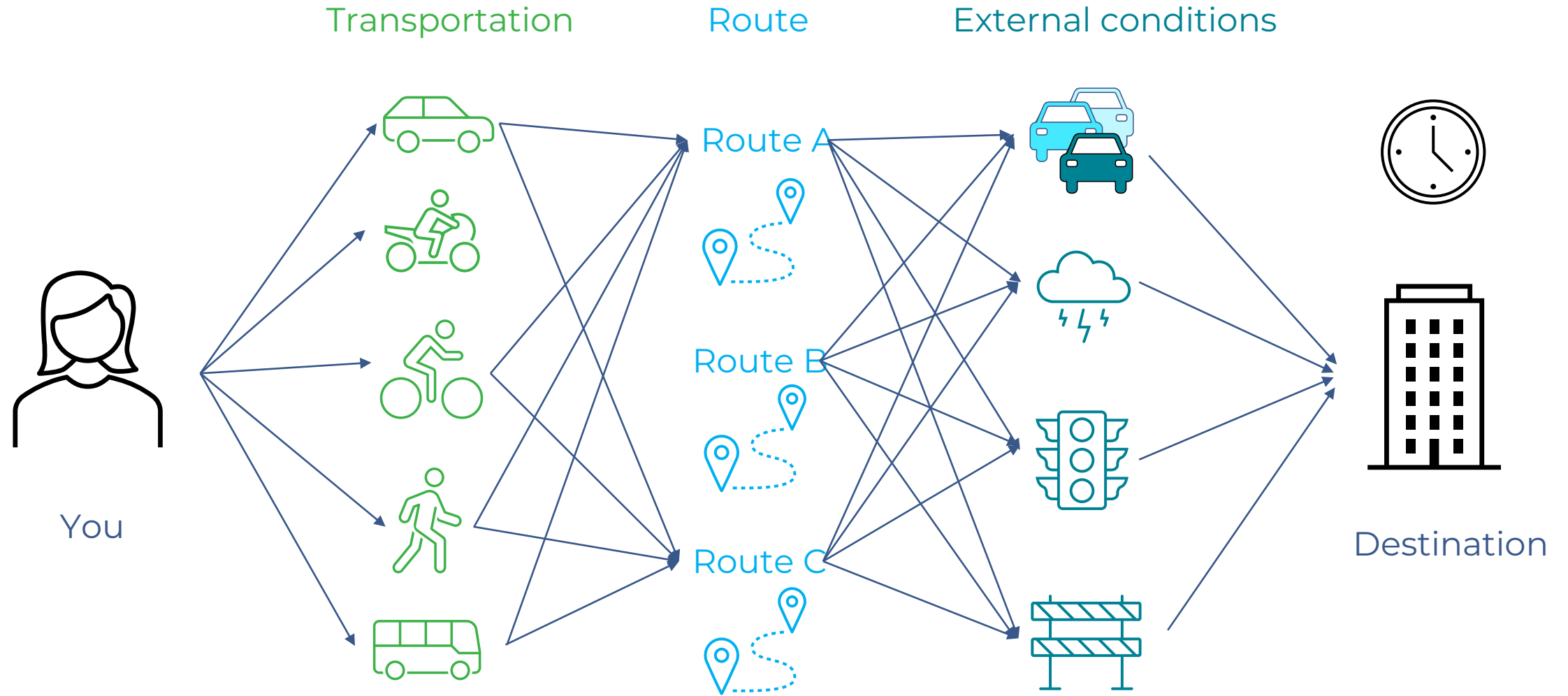
electronic
visualization
laboratory



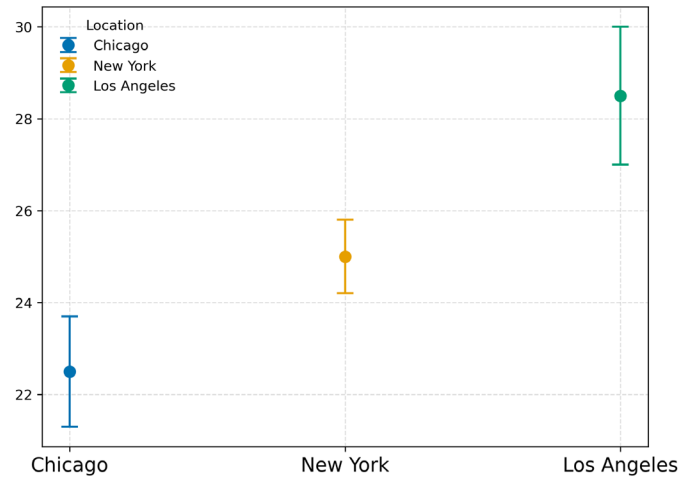
**UNIVERSITY OF
ILLINOIS CHICAGO**

What is uncertainty?

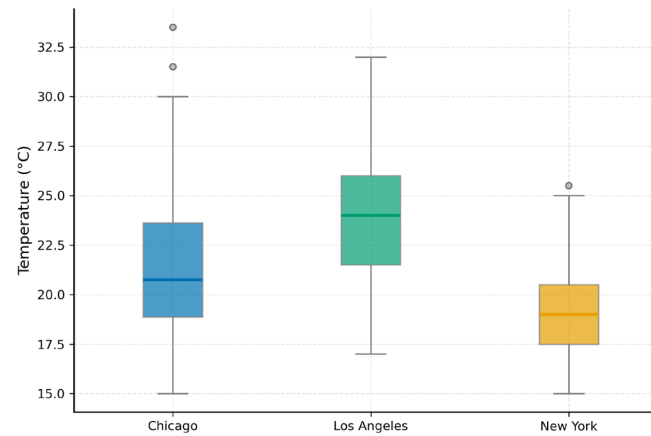
What is uncertainty?



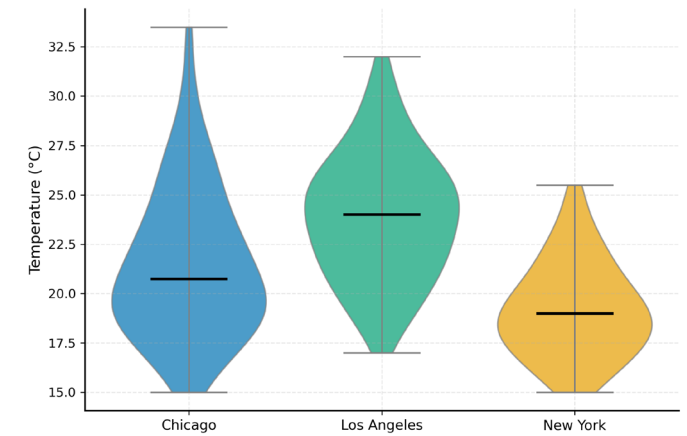
Error Bars



Box Plot

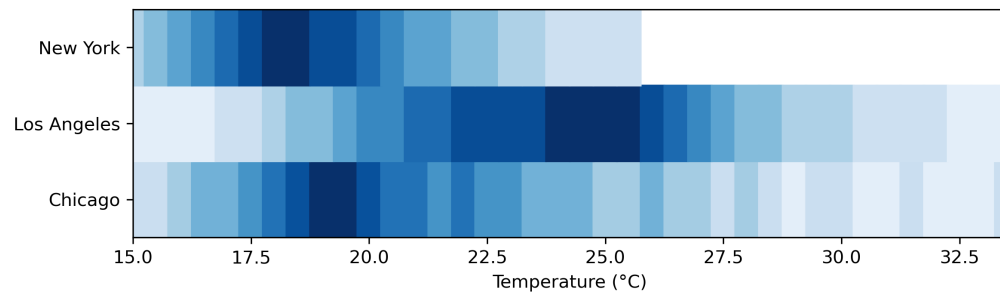


Violin Plot

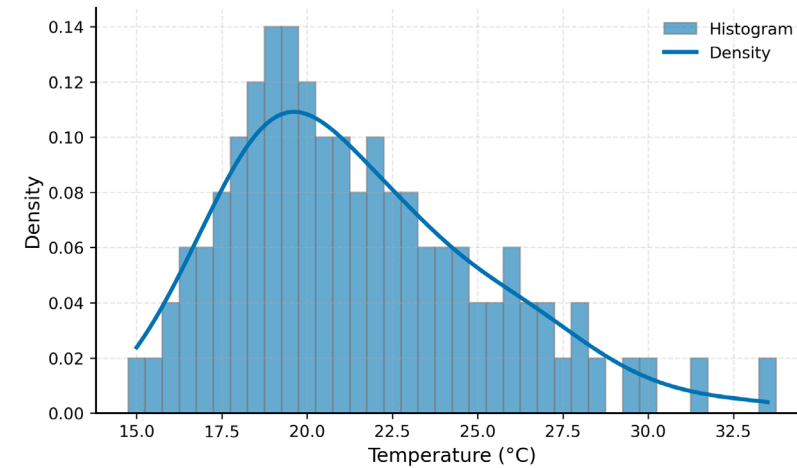


Probability Distribution

Gradient Plot



Temperature Distribution





Commuting time



Stock market



Criminal investigation



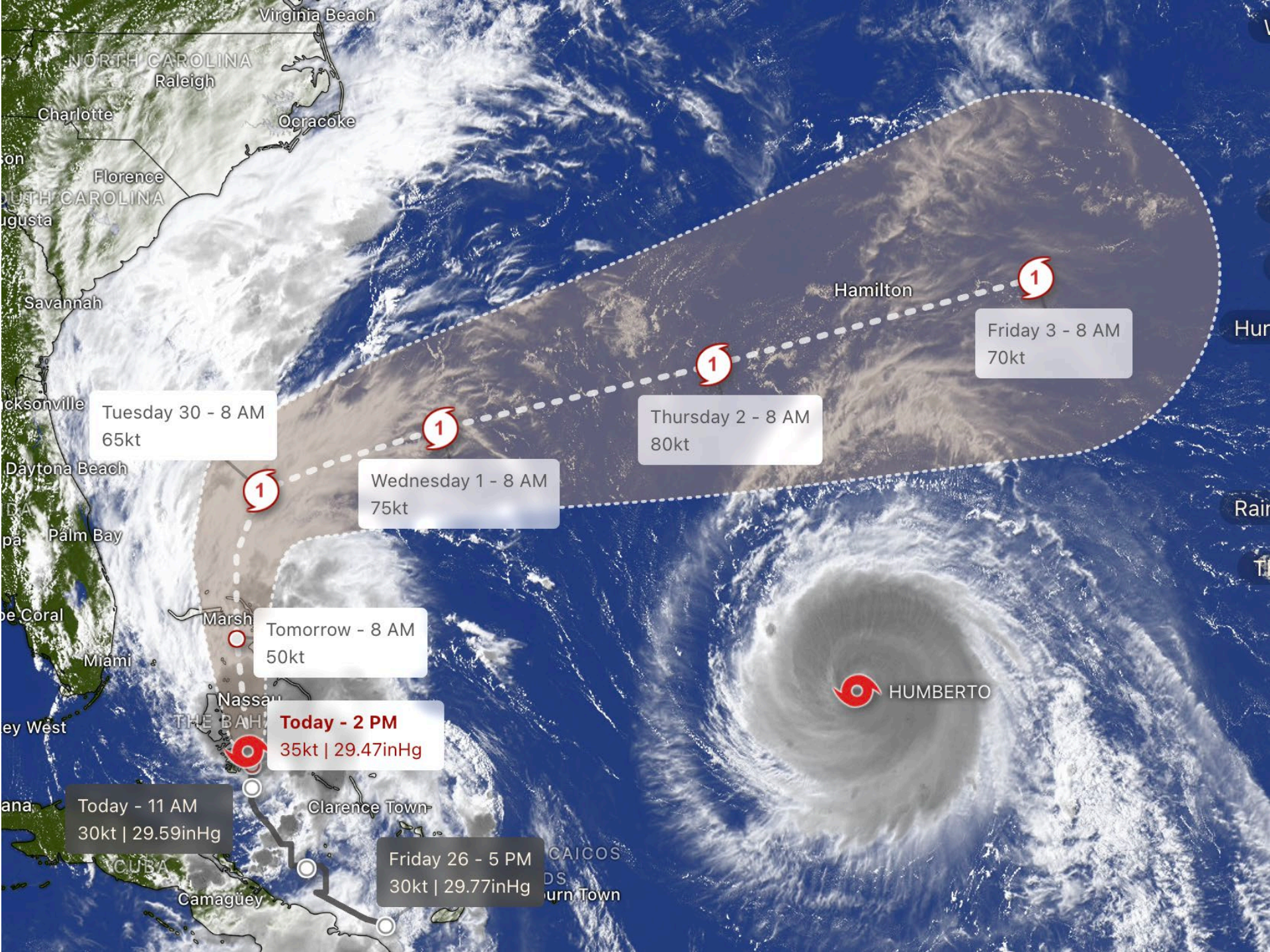
Energy generation



Demand prediction

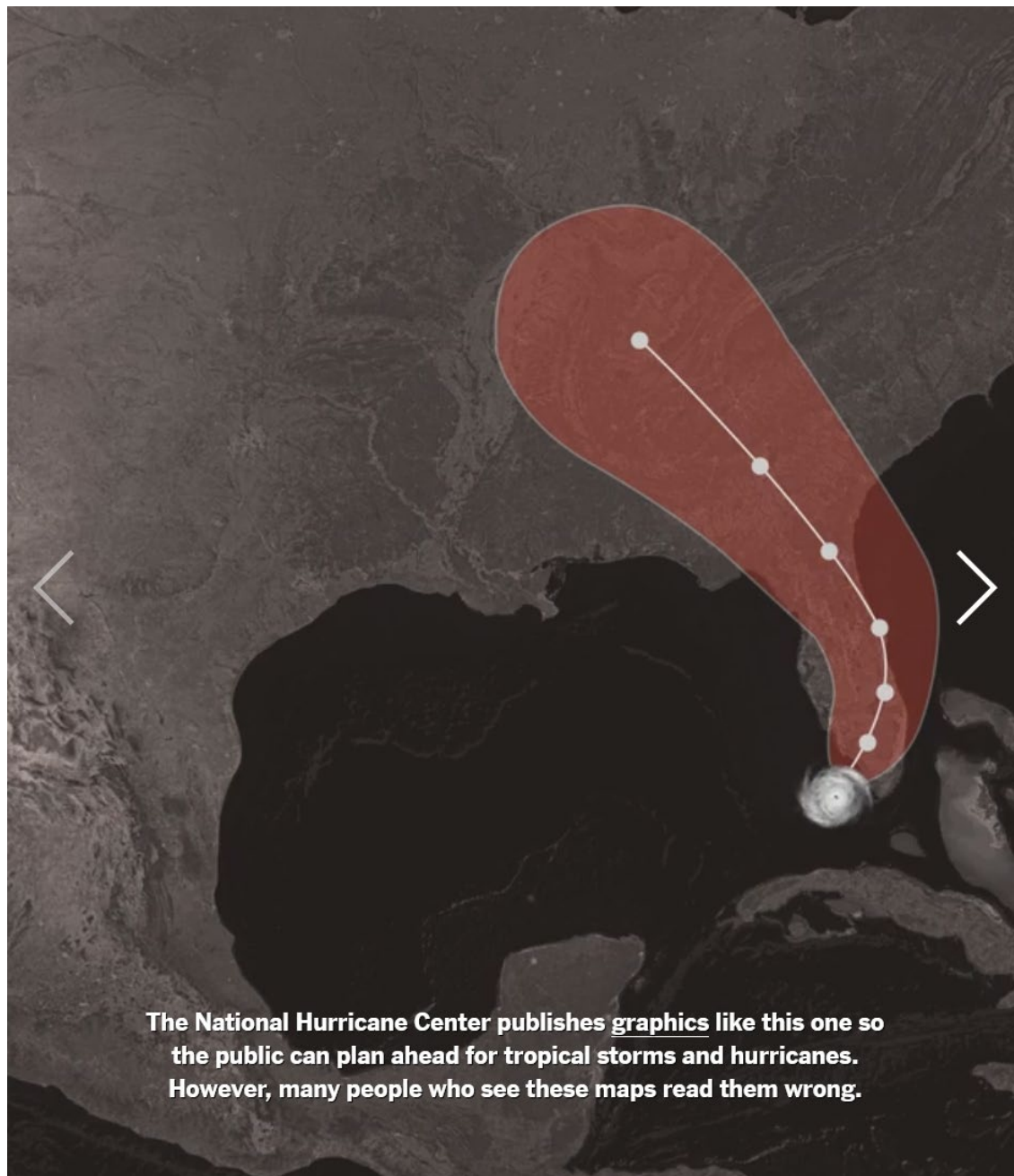


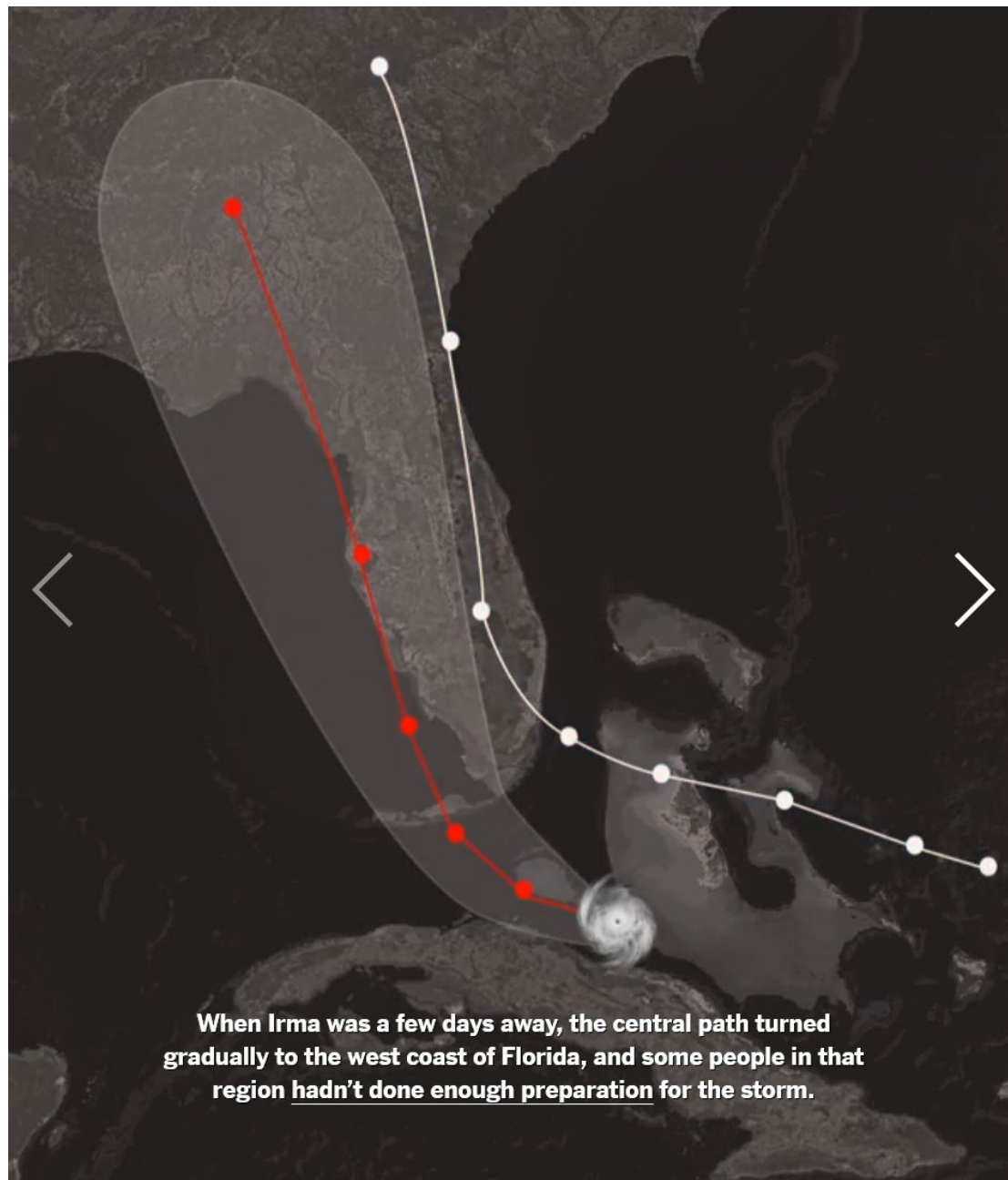
Weather forecasting

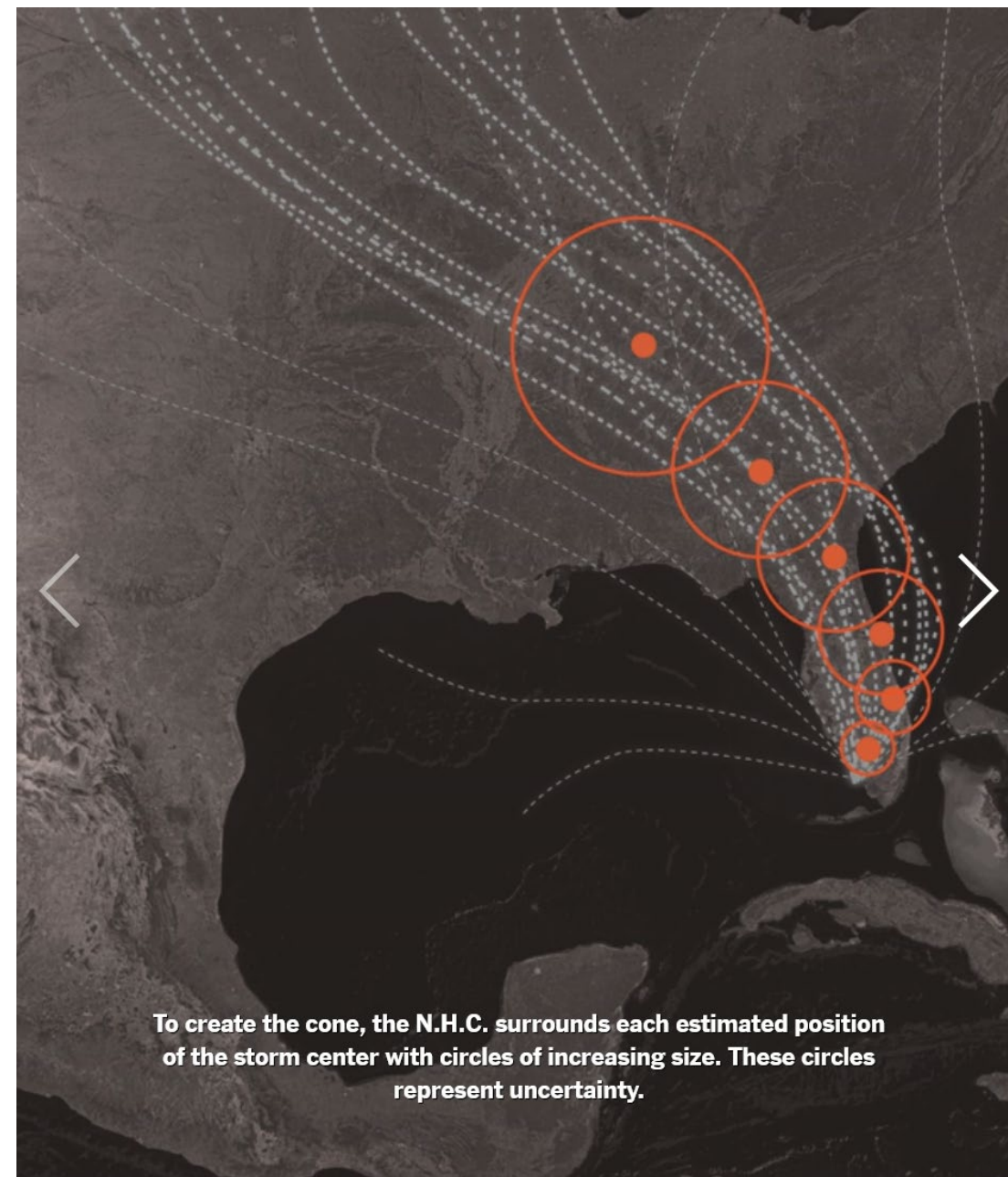


Windy: Wind
map
&
Weather
forecast

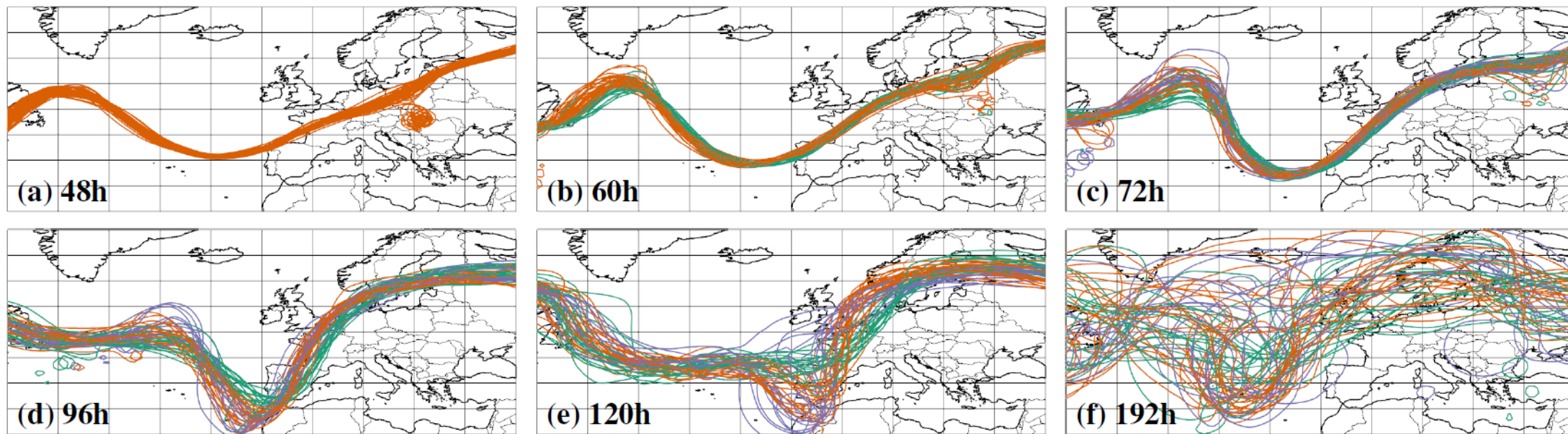
<https://www.windy.com/>



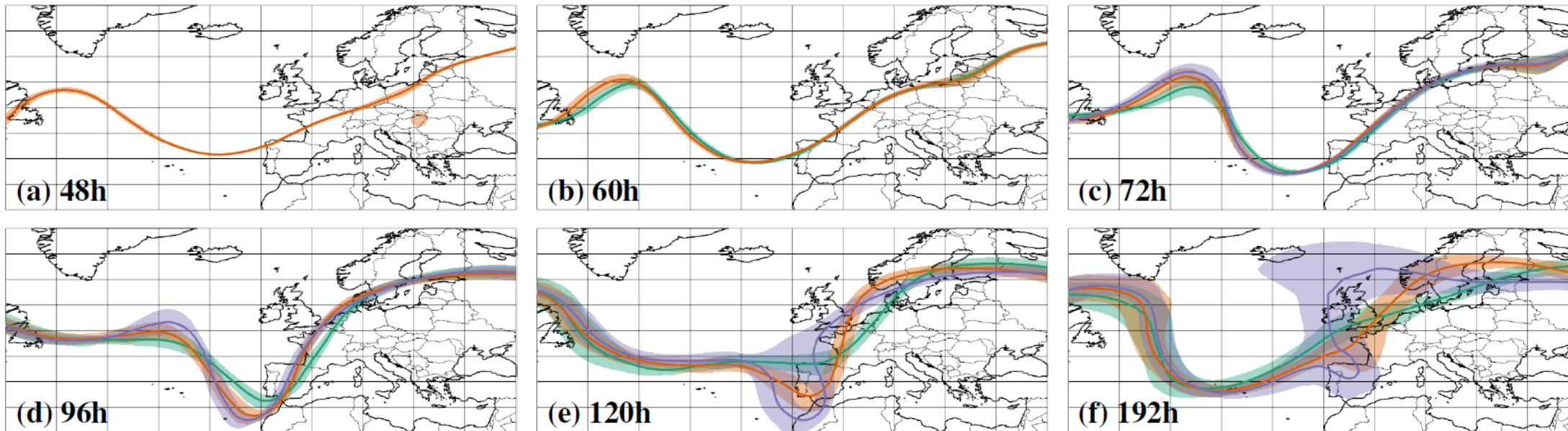




Spaghetti



Contour
boxplots

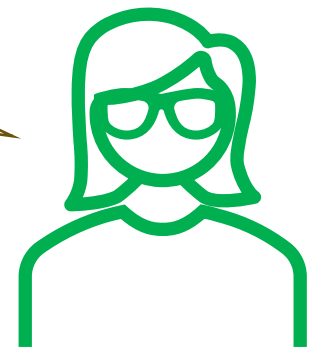


We have an **ensemble**, and need to answer:
How probable is a storm in this area?
Which setup gives the best results?
How do different setups affect the results?



Climate Expert

How can I
help?



Data Visualization Expert

I am
used to
static
images

Creating some
interactive maps and
charts.

Creating a dashboard.

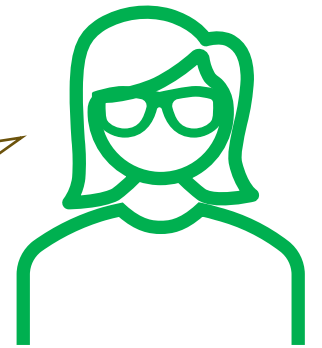
A common
dashboard may not
be enough.

Plotting variables in a
3D map?

Show me your data.
Tell me about your pipeline.
How do you usually work?



Climate Expert



Data Visualization Expert



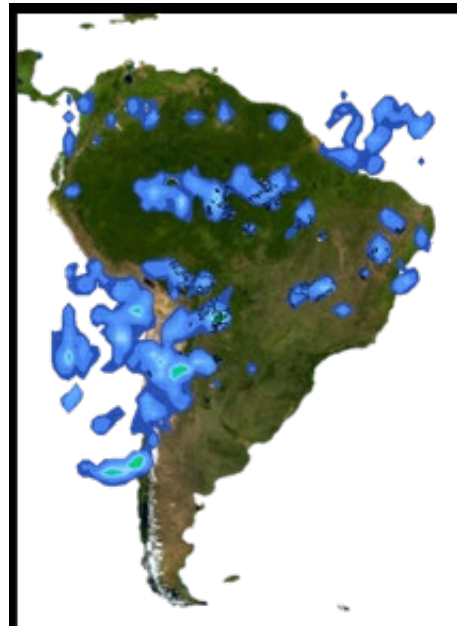
How to model the atmosphere?

Start

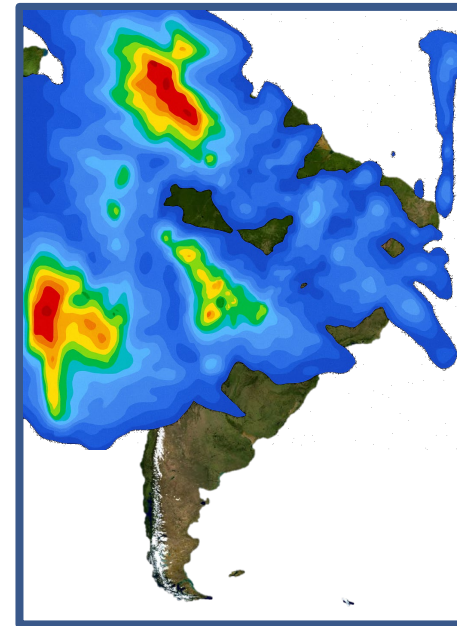
End



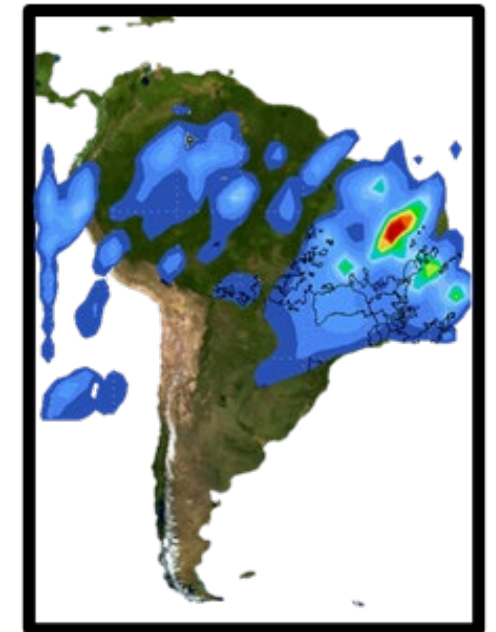
Time step 1



Time step 2



Time step 3



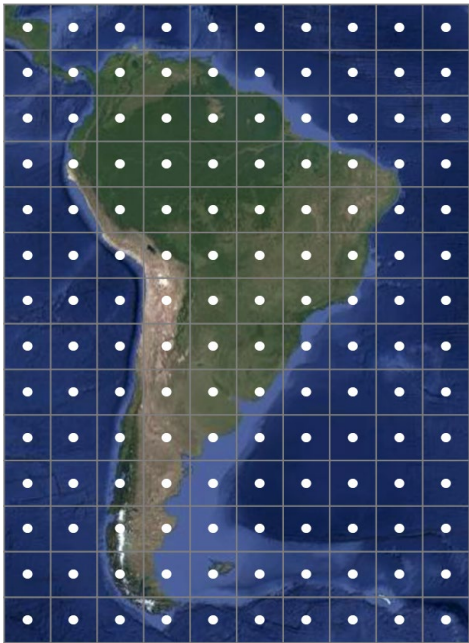
Time step 4



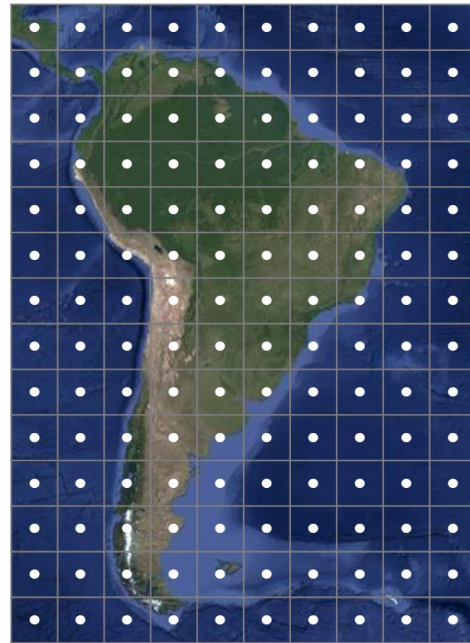
How to model the atmosphere?

Start

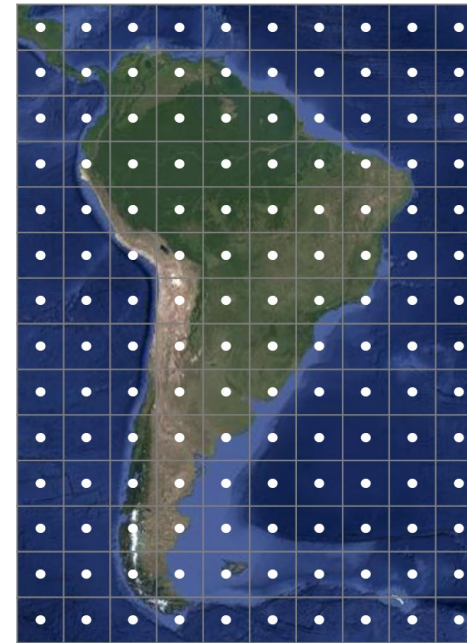
End



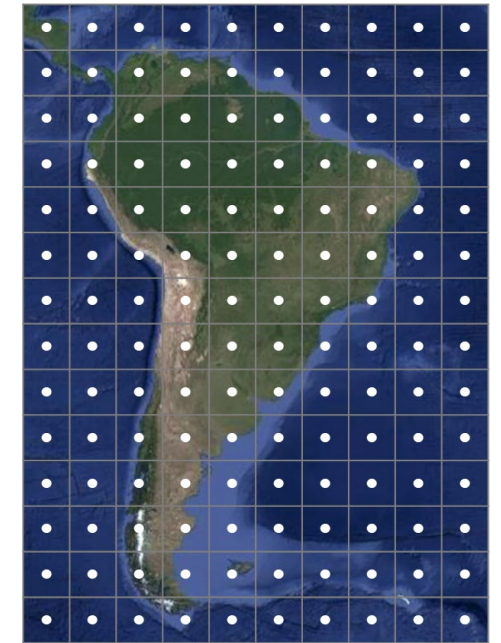
Time step 1



Time step 2



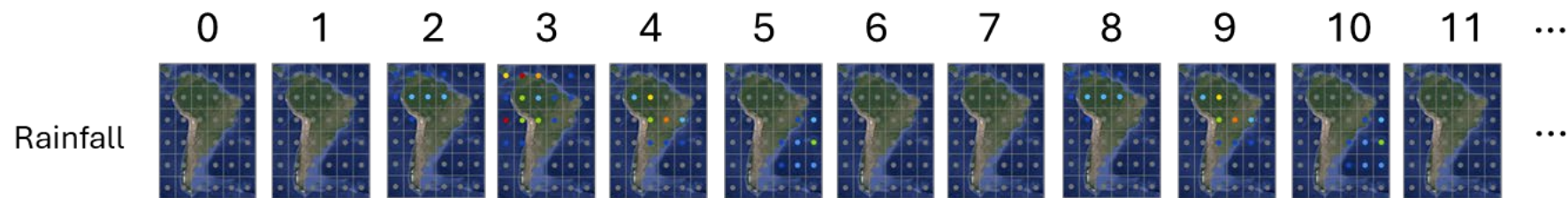
Time step 3



Time step 4

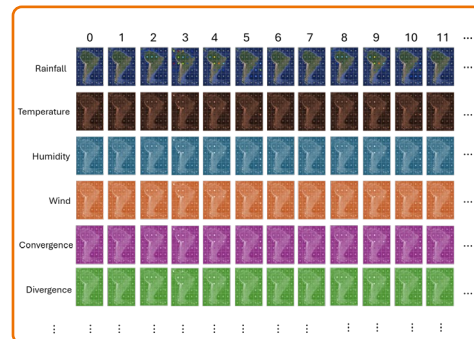


VARIABLES

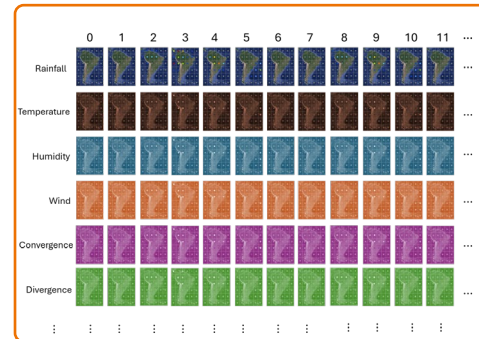


RUNS

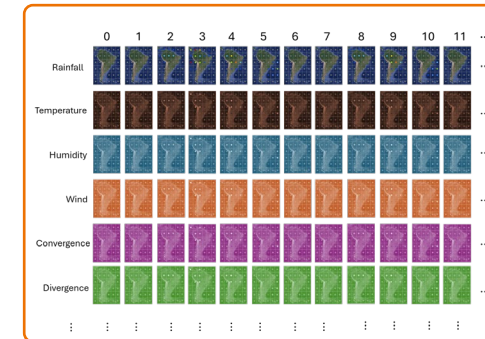
Run 1



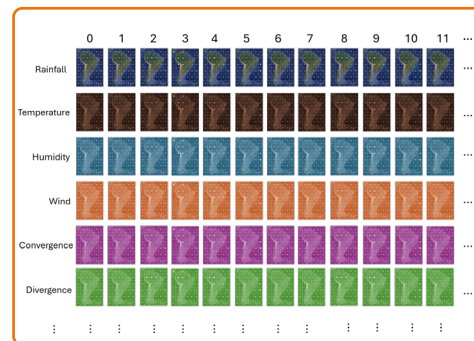
Run 2



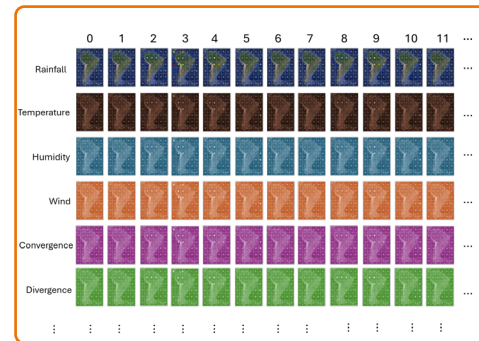
Run 3



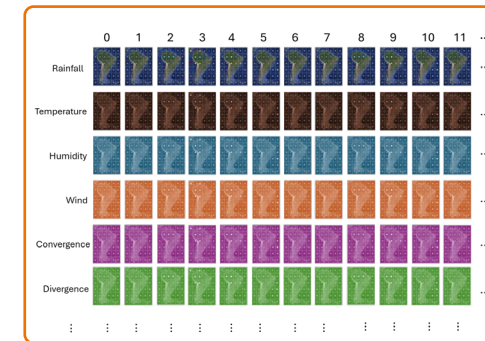
Run 4



Run 5



Run 6

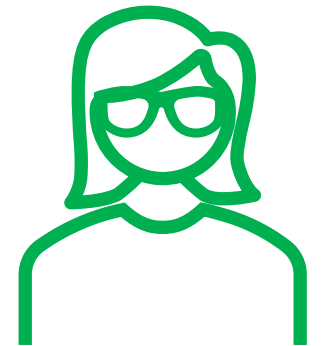


...

Tell me more.



Climate Expert

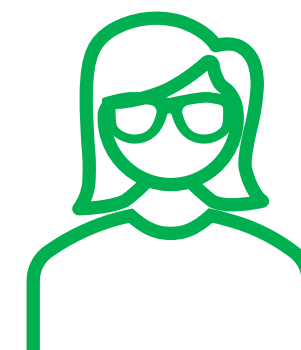
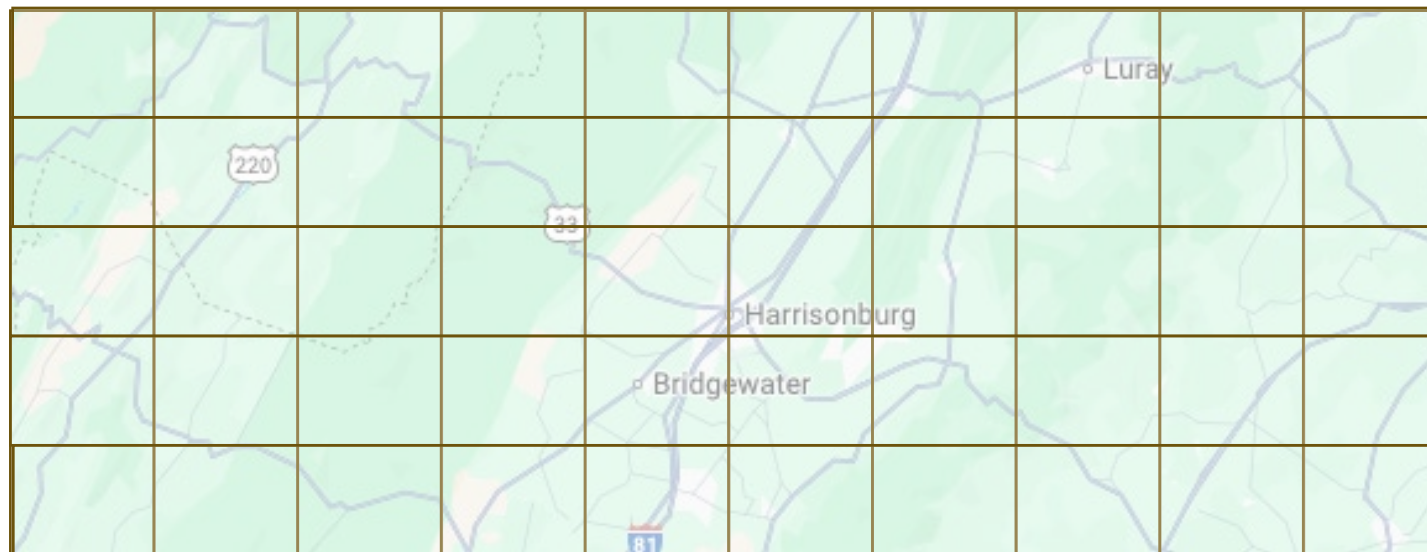


Data Visualization Expert

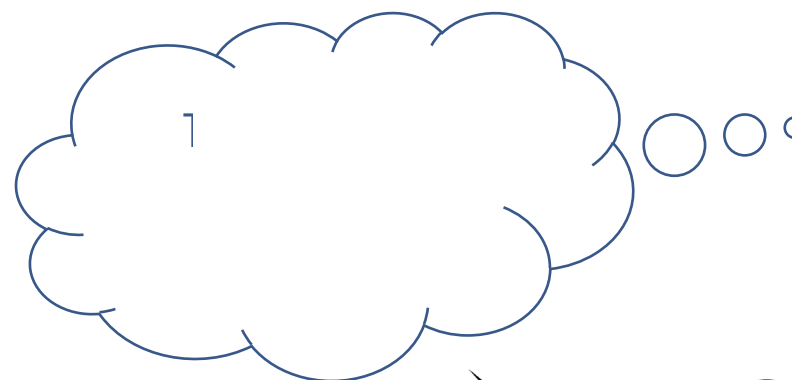


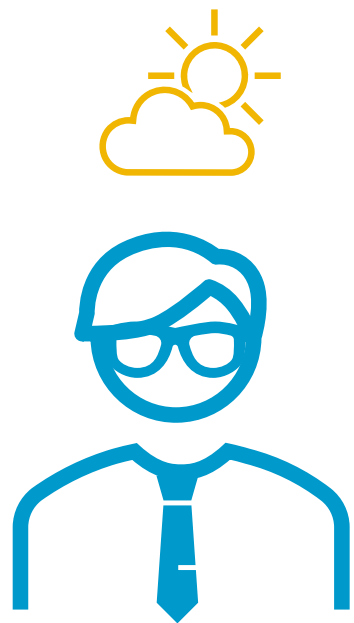
Climate Expert

10 x 5
points

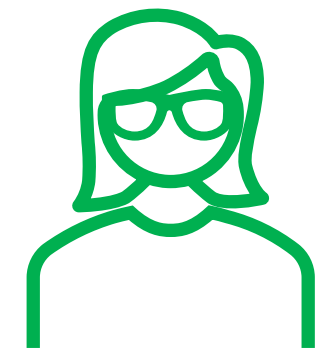
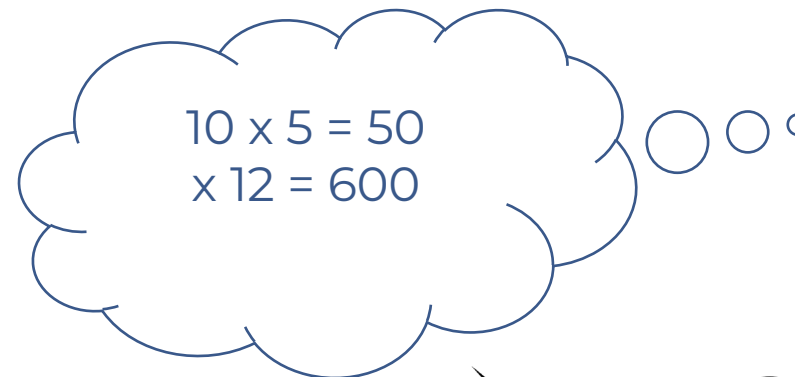
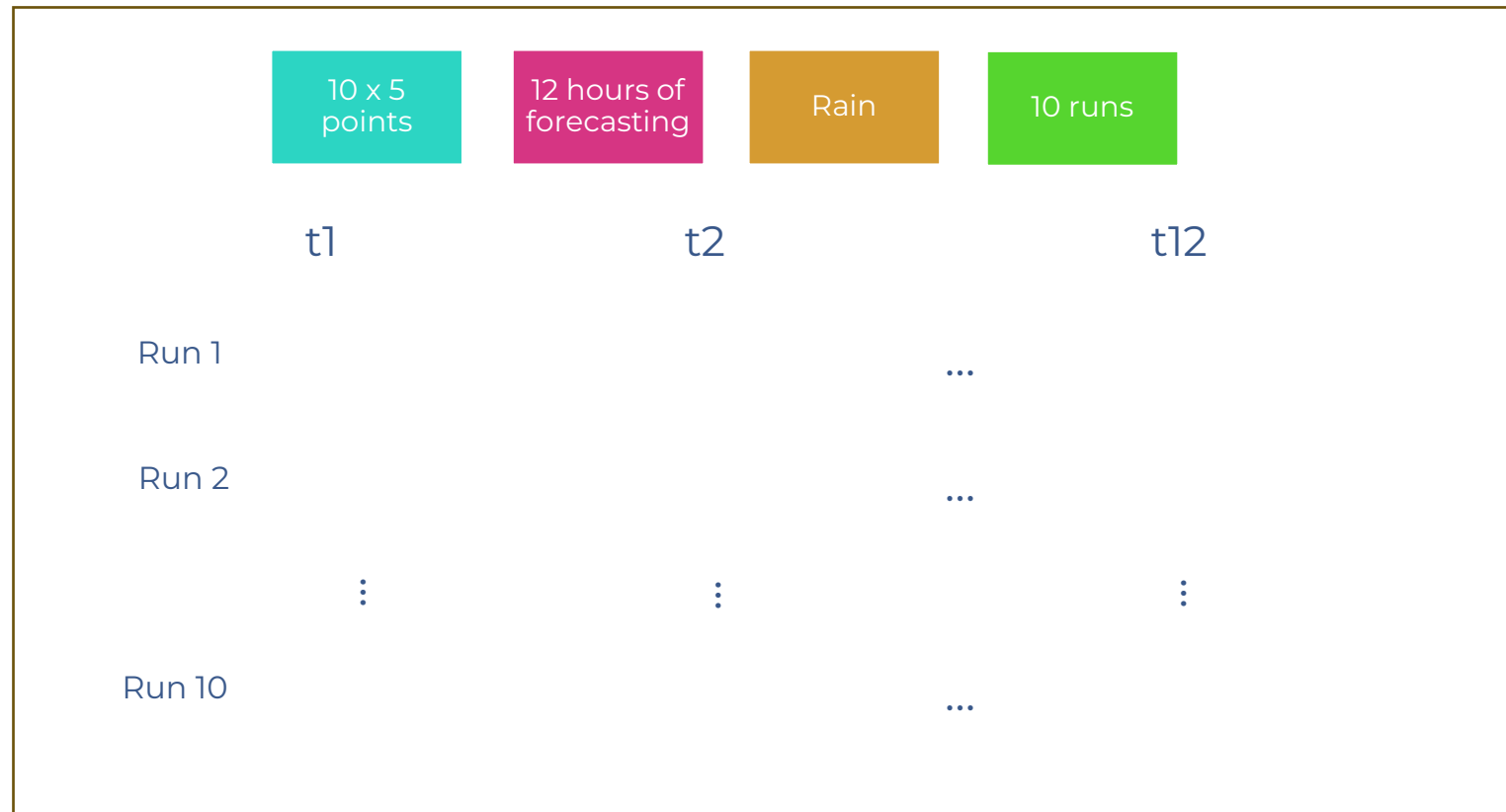


Data Visualization Expert





Climate Expert



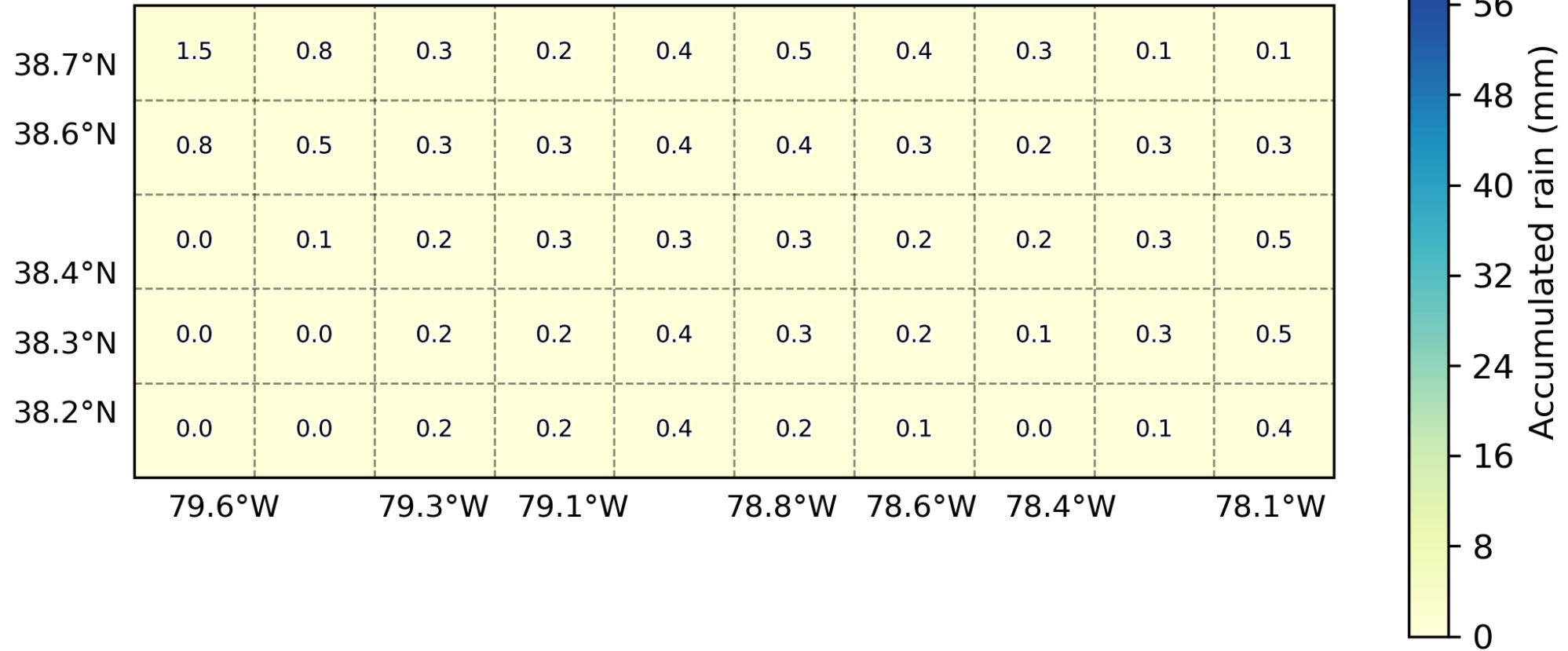
Data Visualization Expert

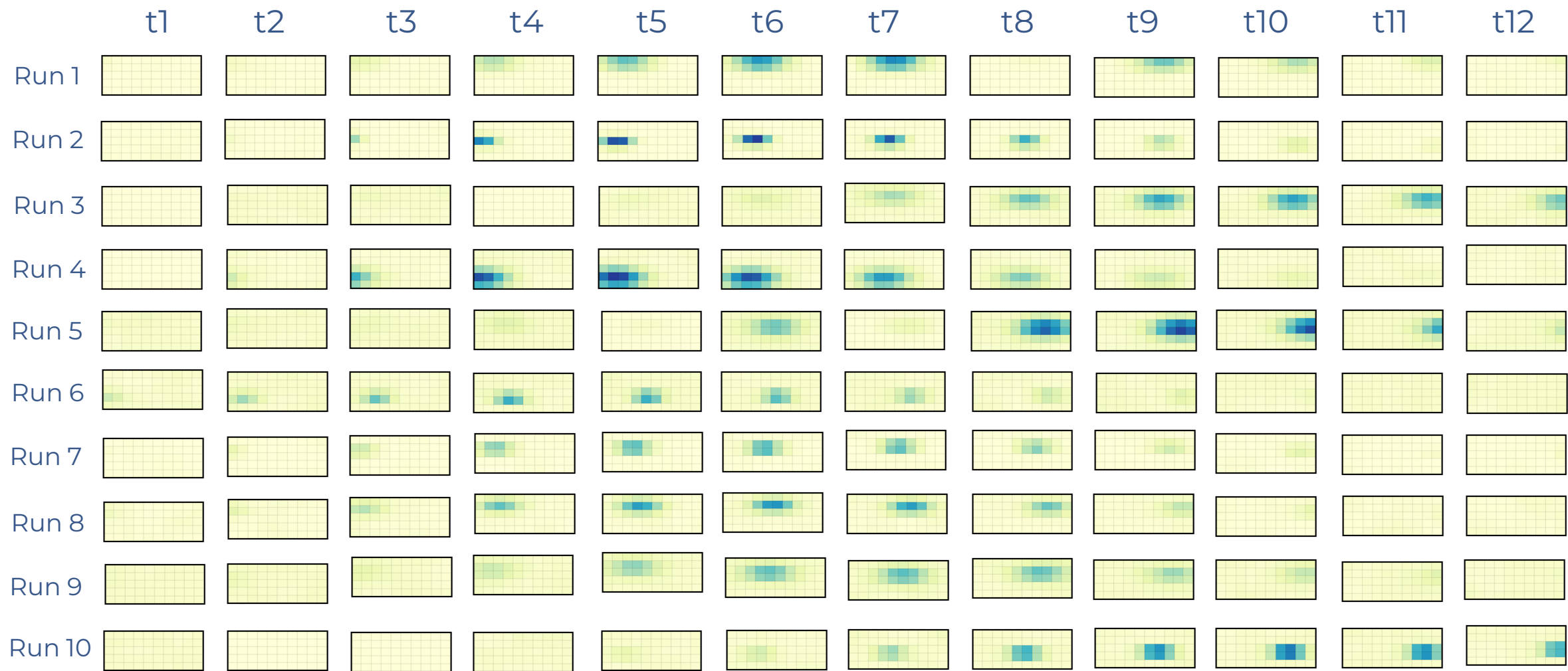
What's the probability of rain?

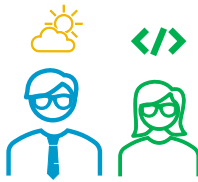
How much? Where?



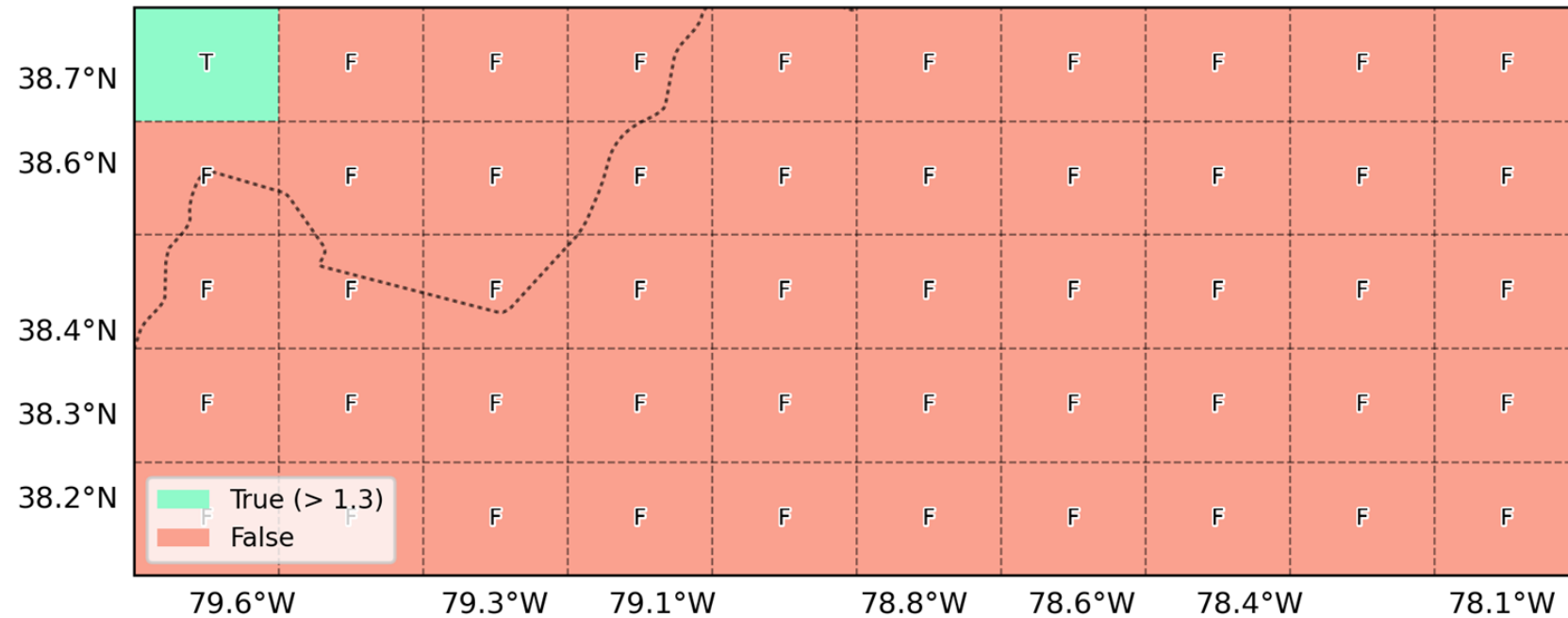
Run 1 t1





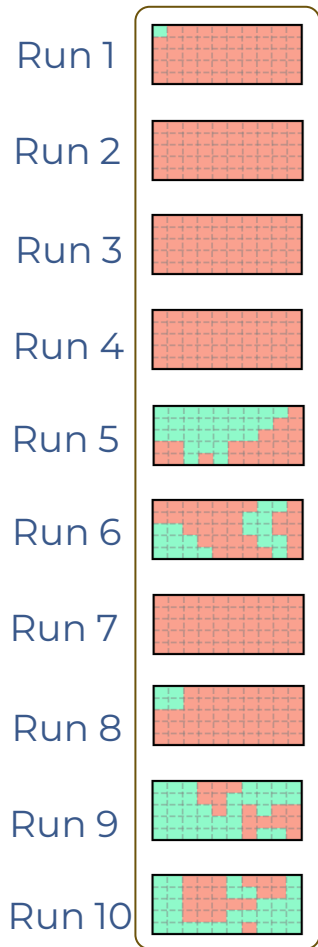


Run 1 t1

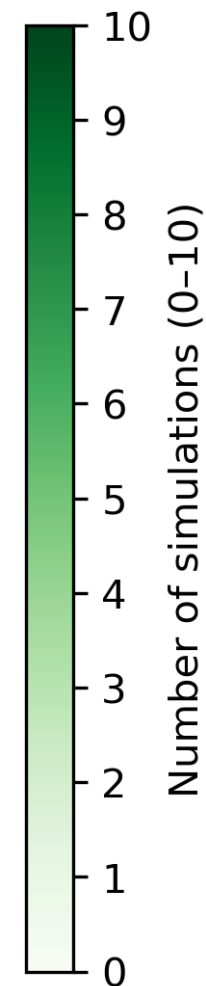
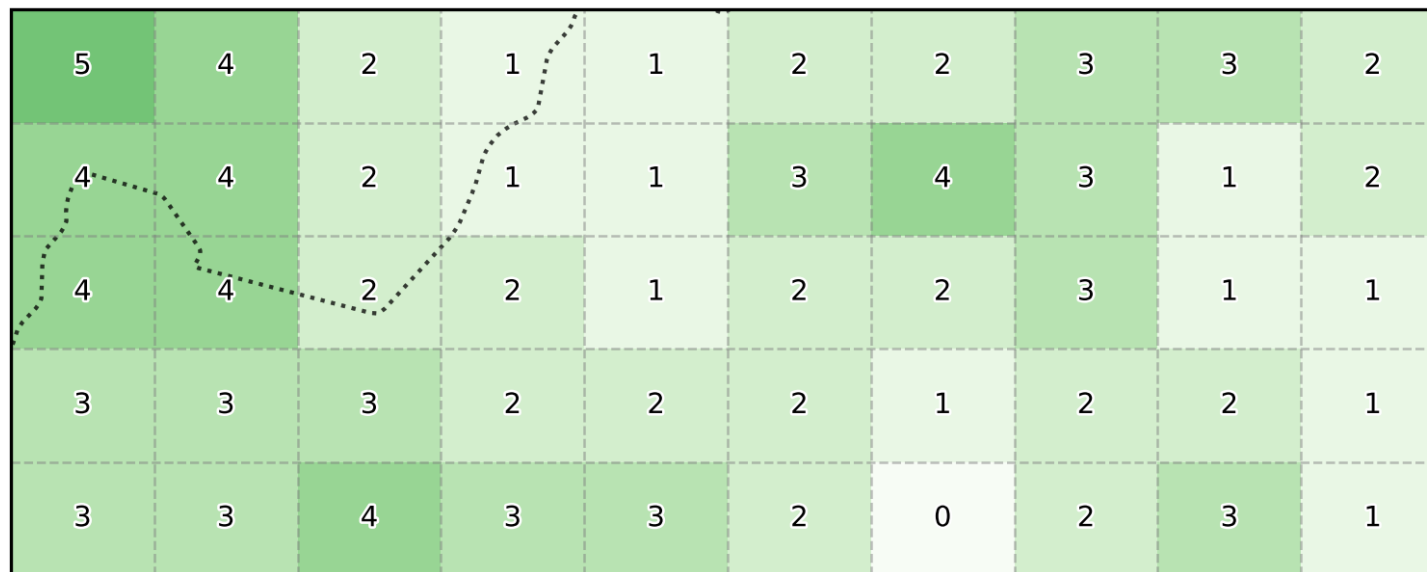


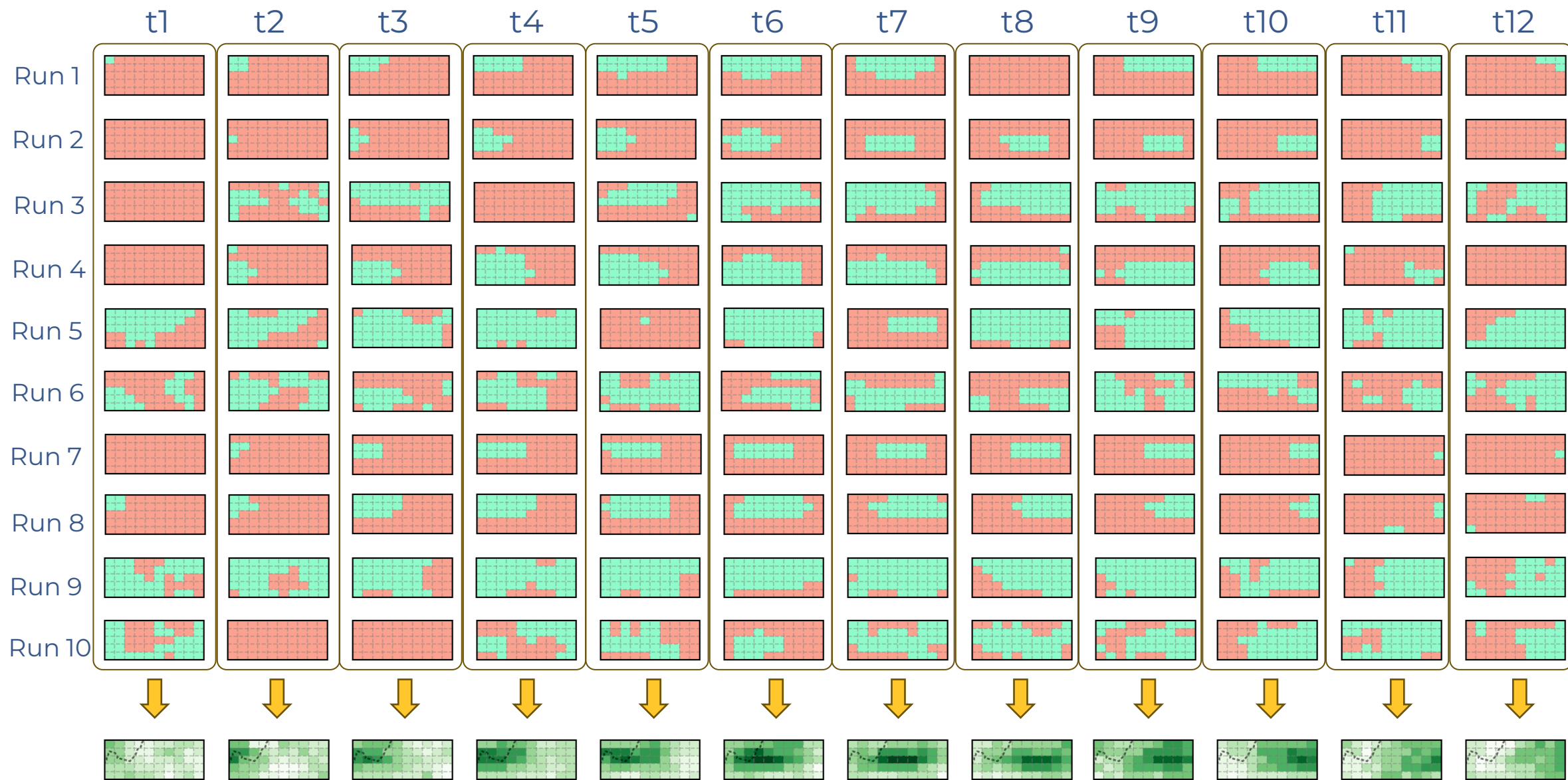
Condition: rain > 1.3 mm/h

True/False rain > 1.3 mm/h



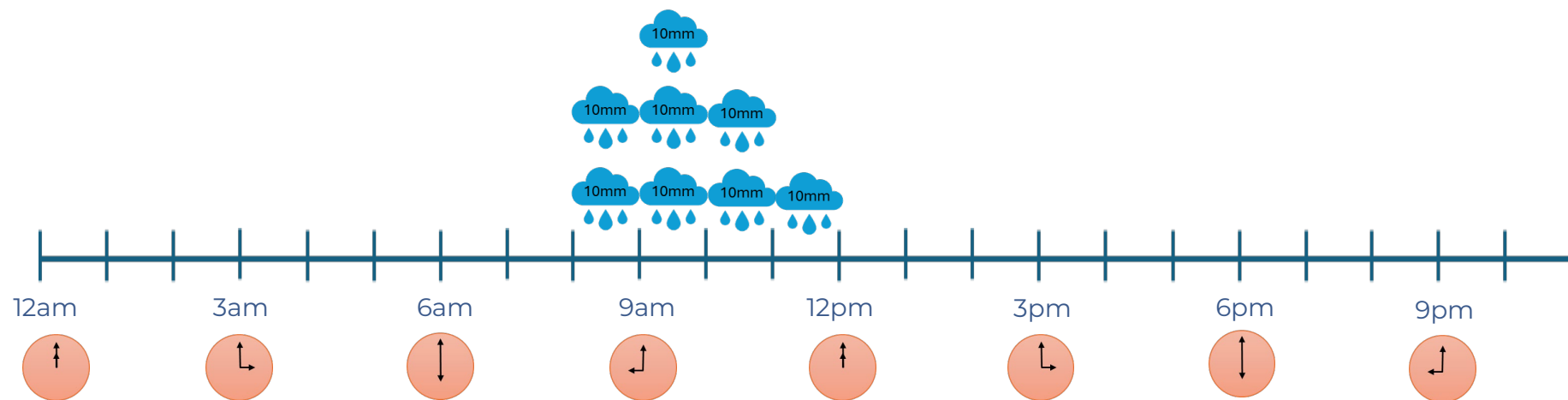
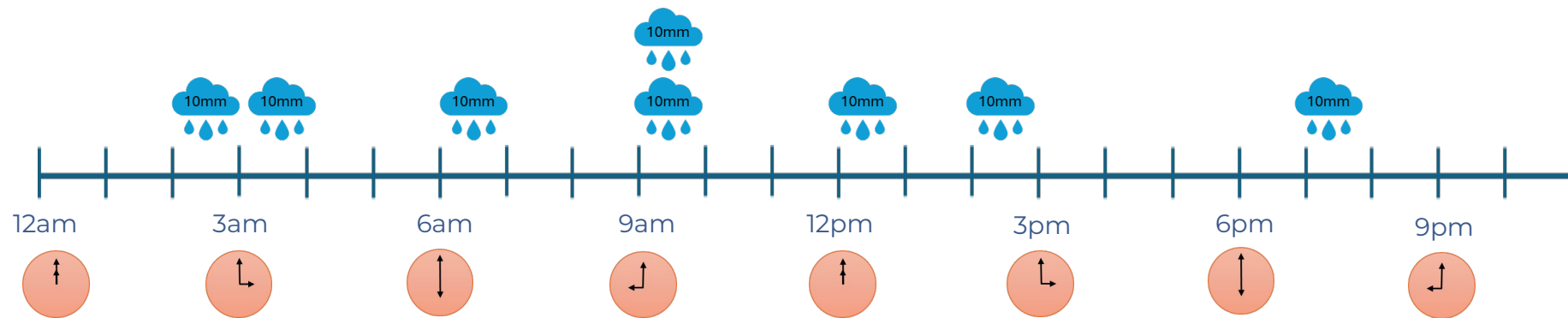
Number of runs that predicted rain > 1.3 mm/h → t1



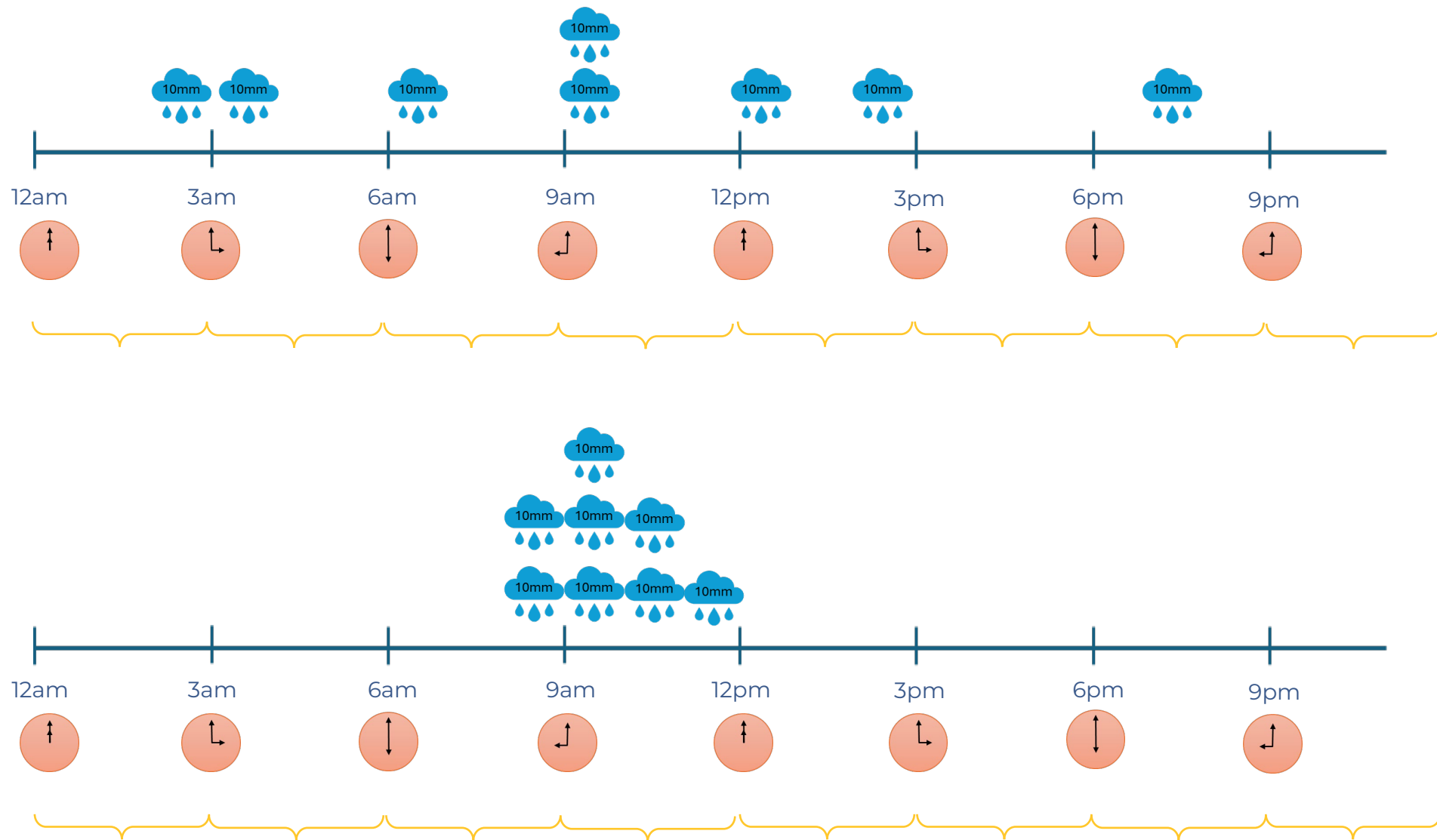


Number of runs that predicted rain > 1.3 mm/h in each time step

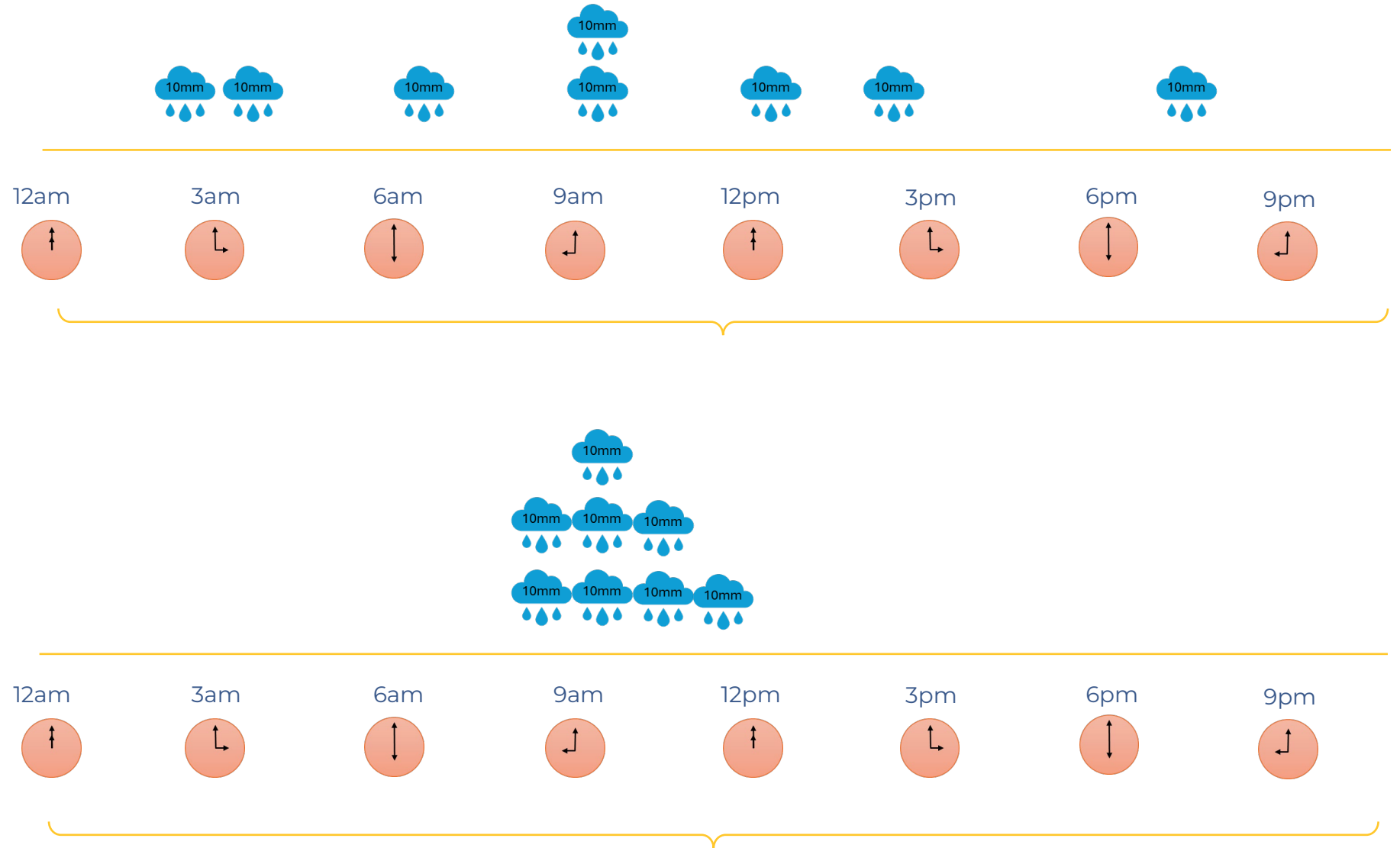
Variable Rain



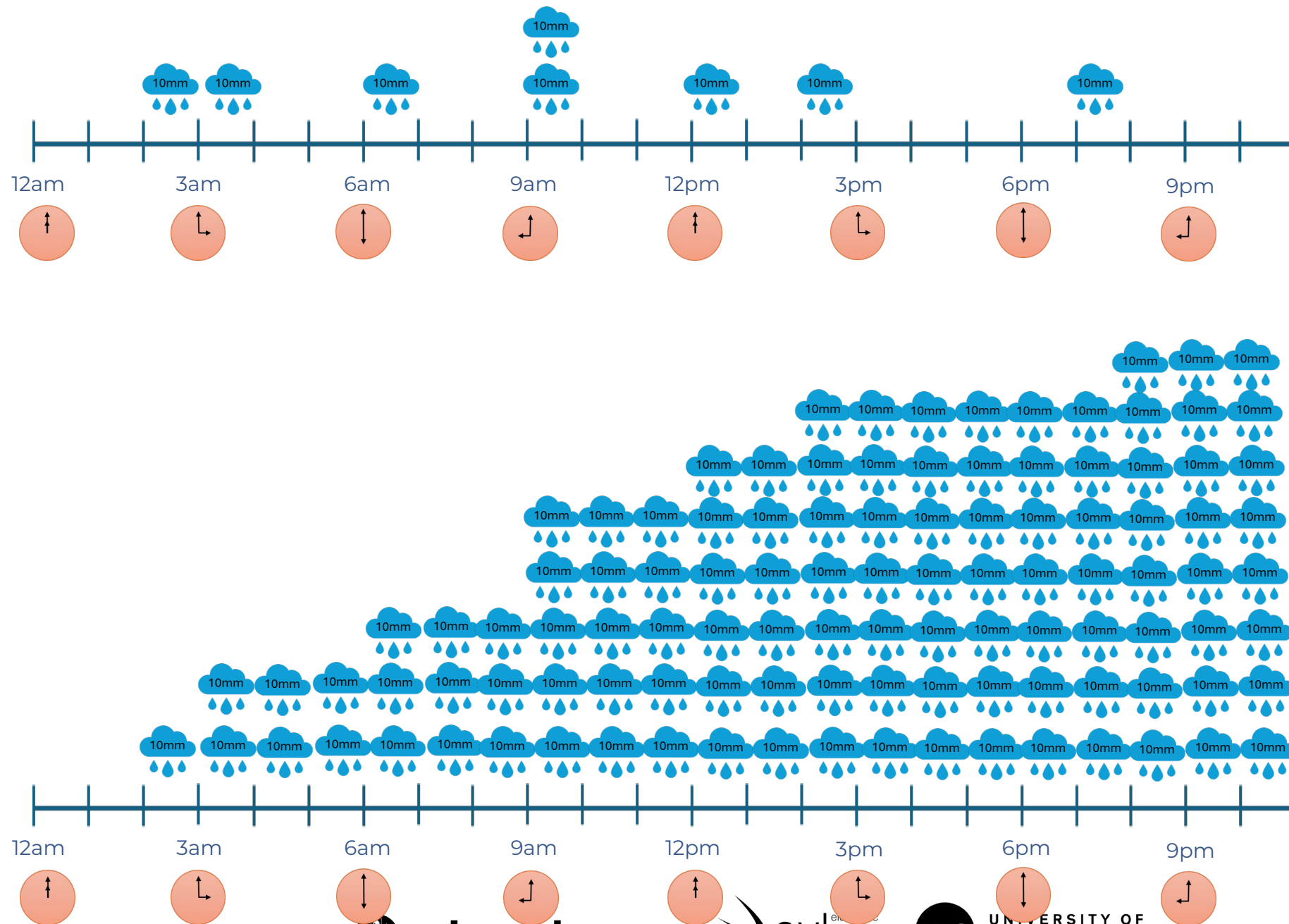
Variable Rain

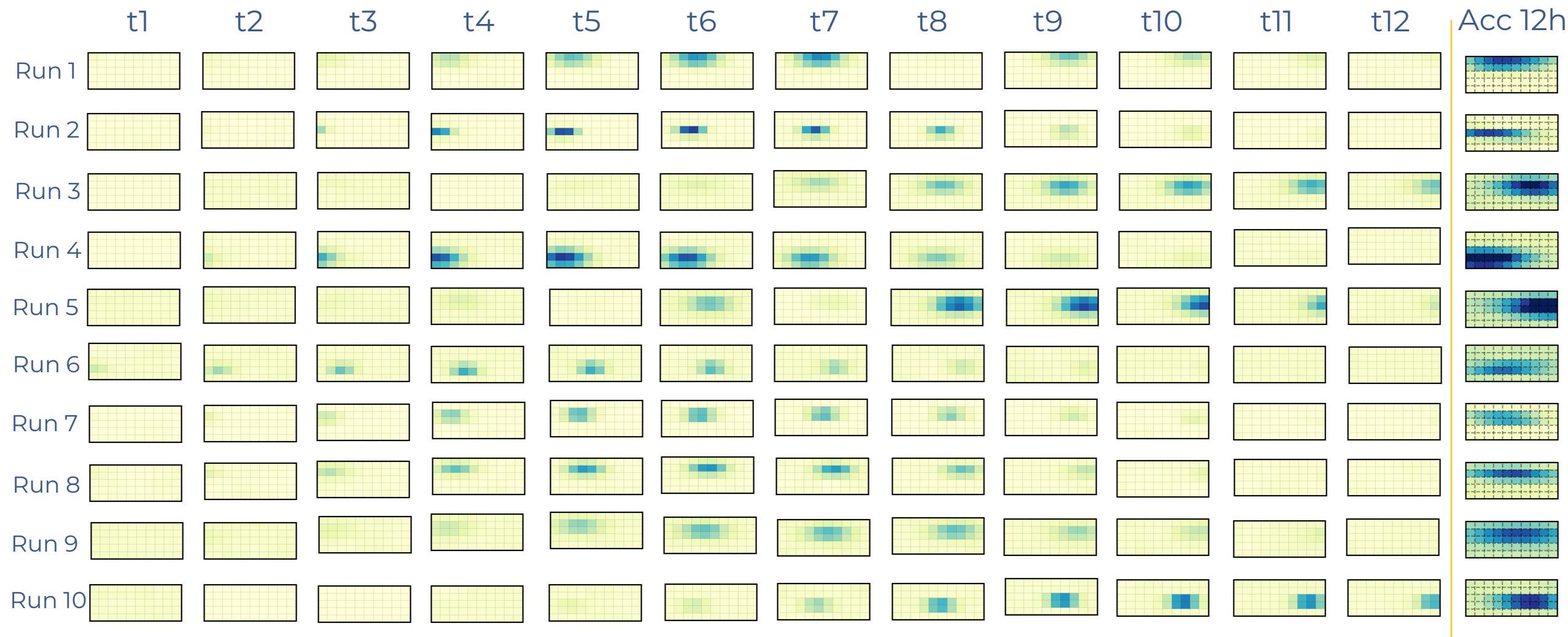


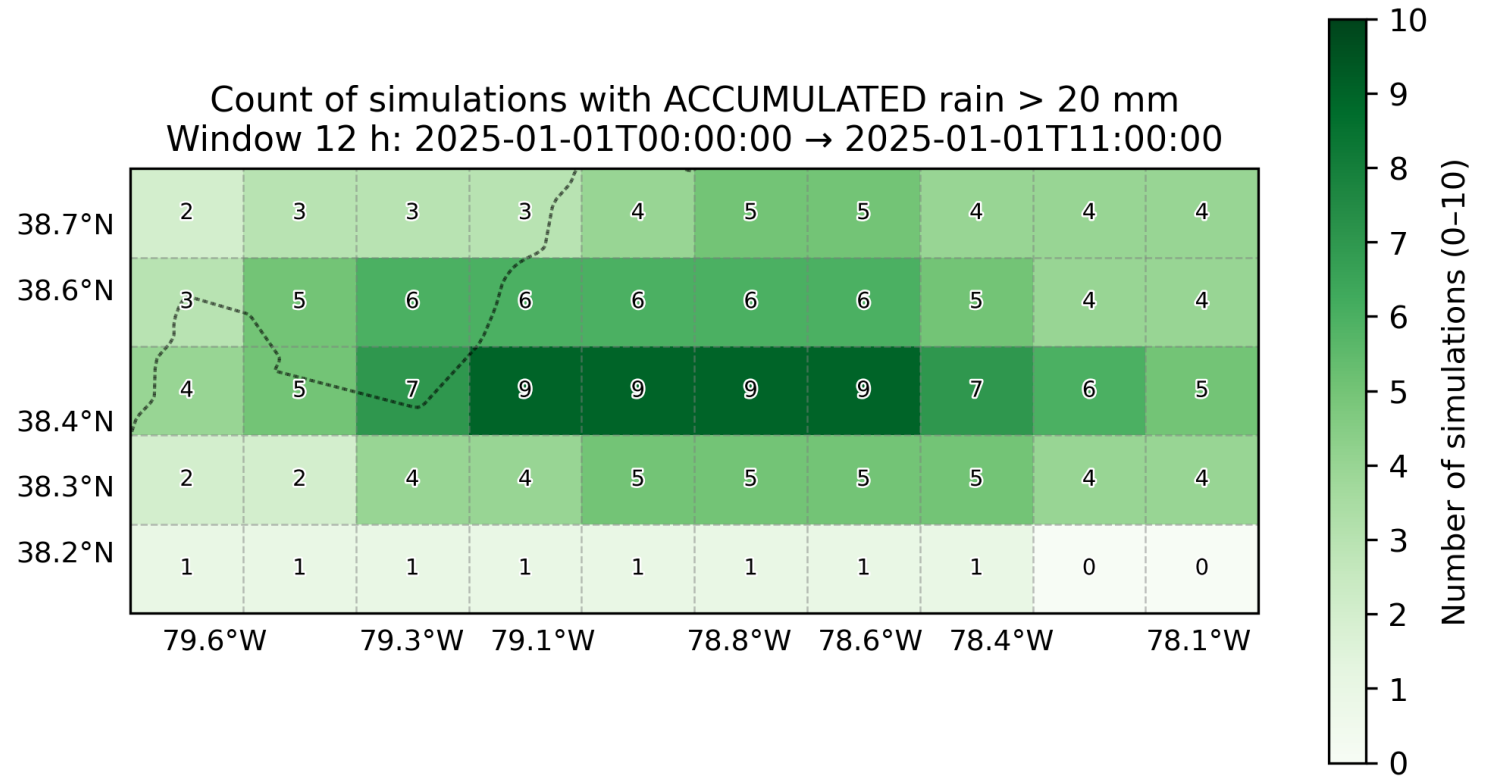
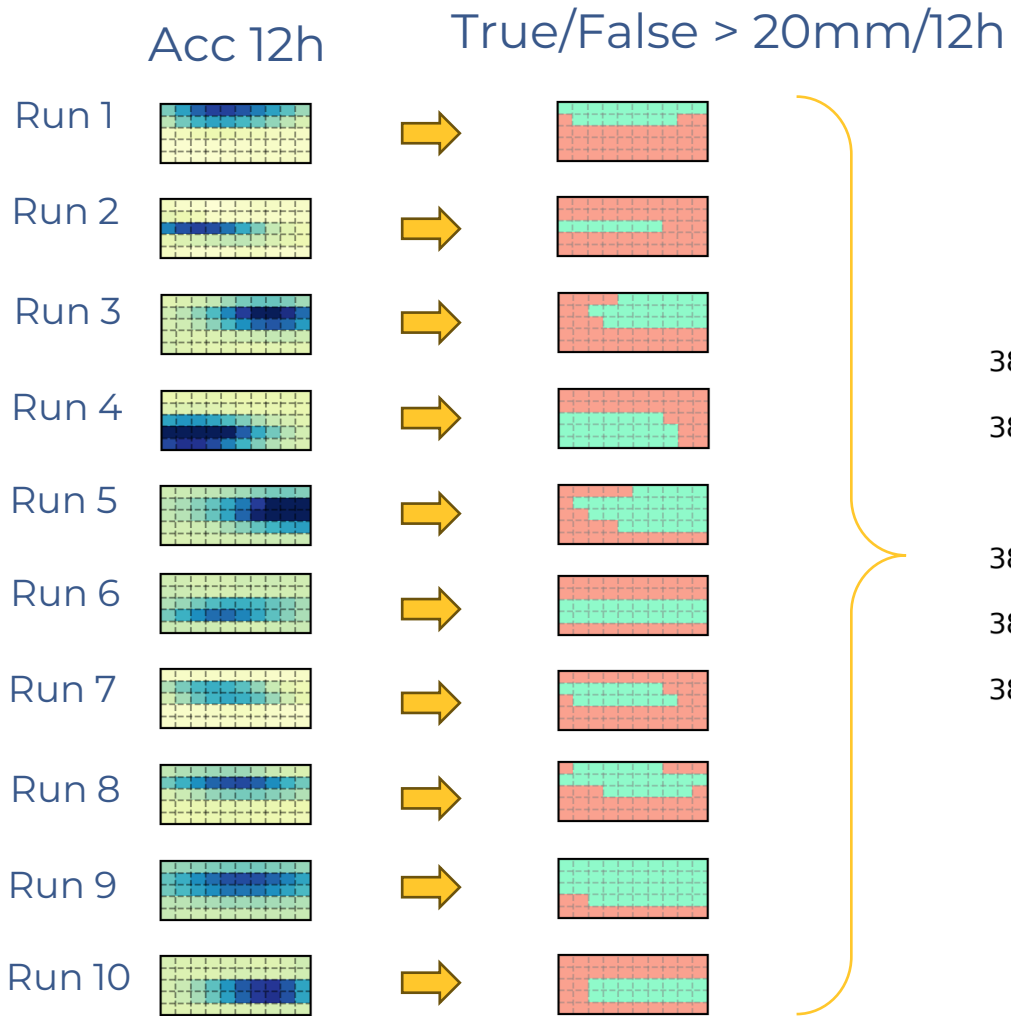
Variable Rain



Variable Rain

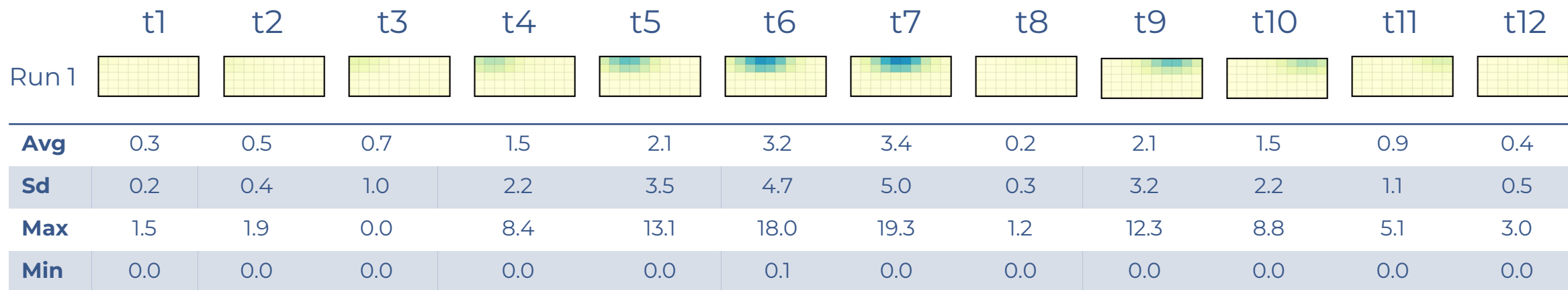




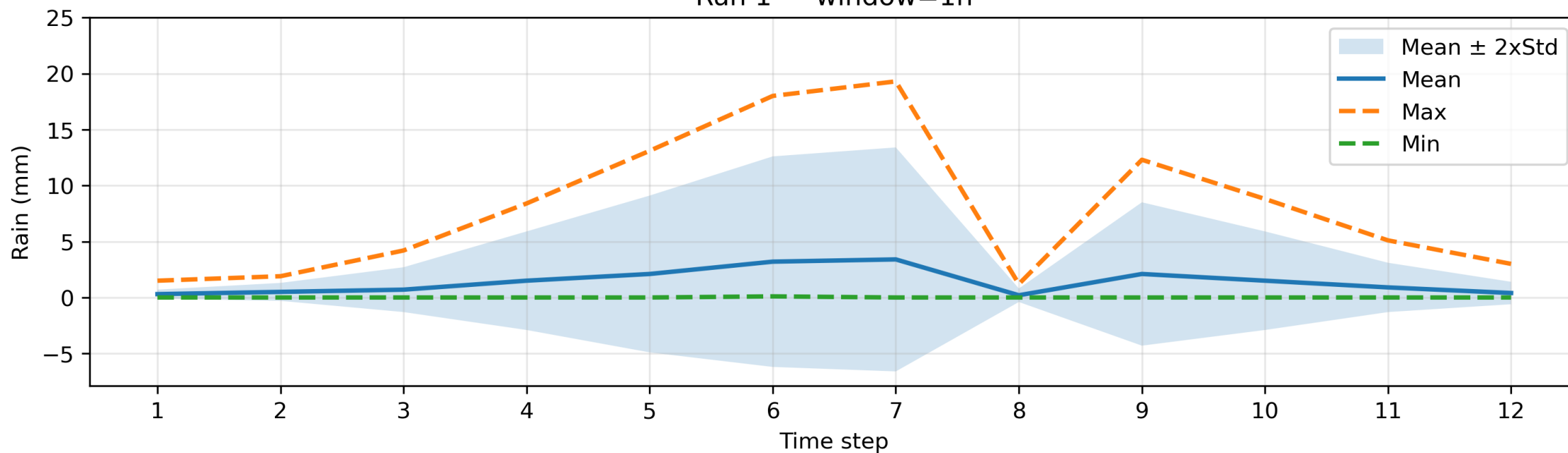


Which simulations had the highest predicted rainfall?

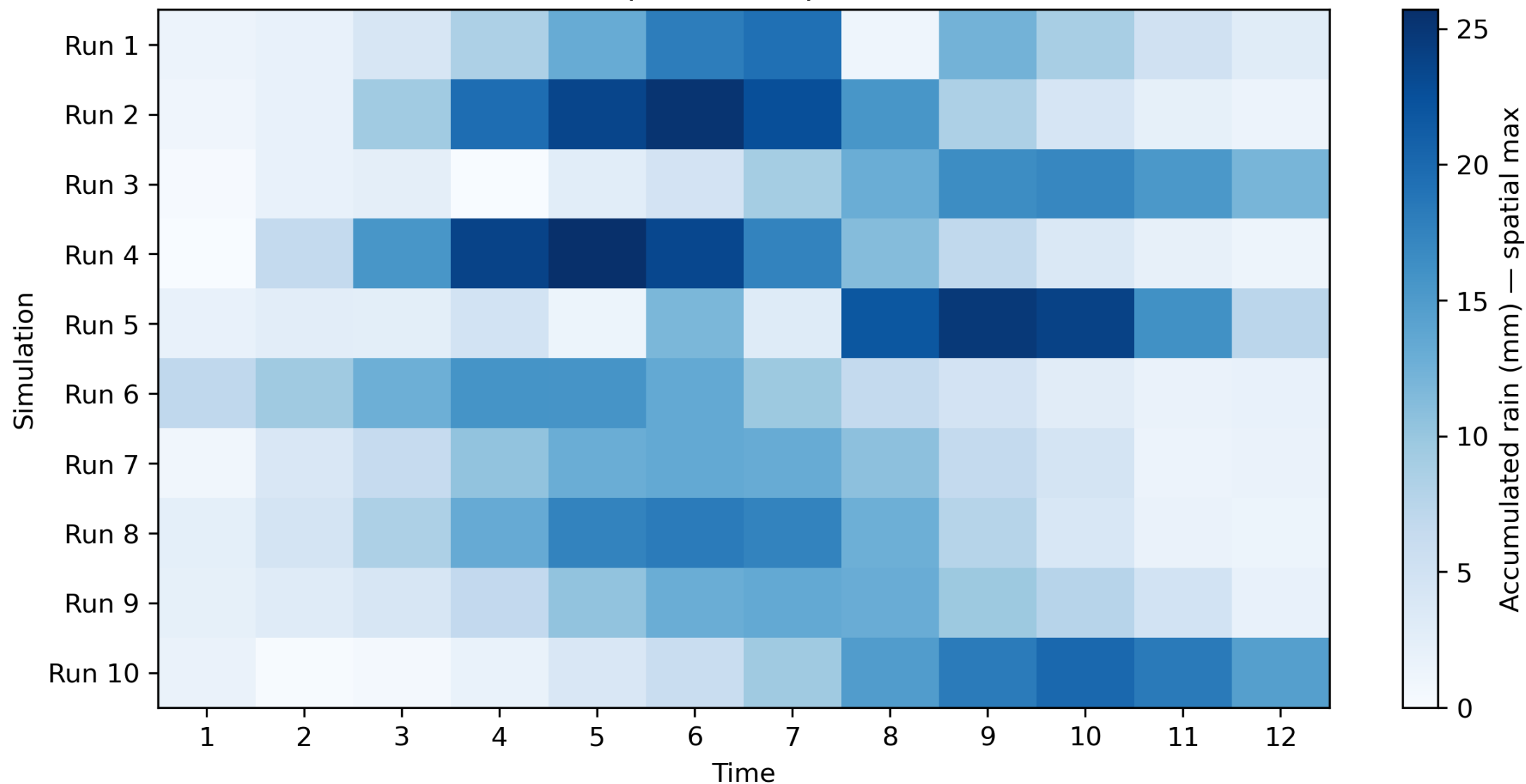
How much? Where?

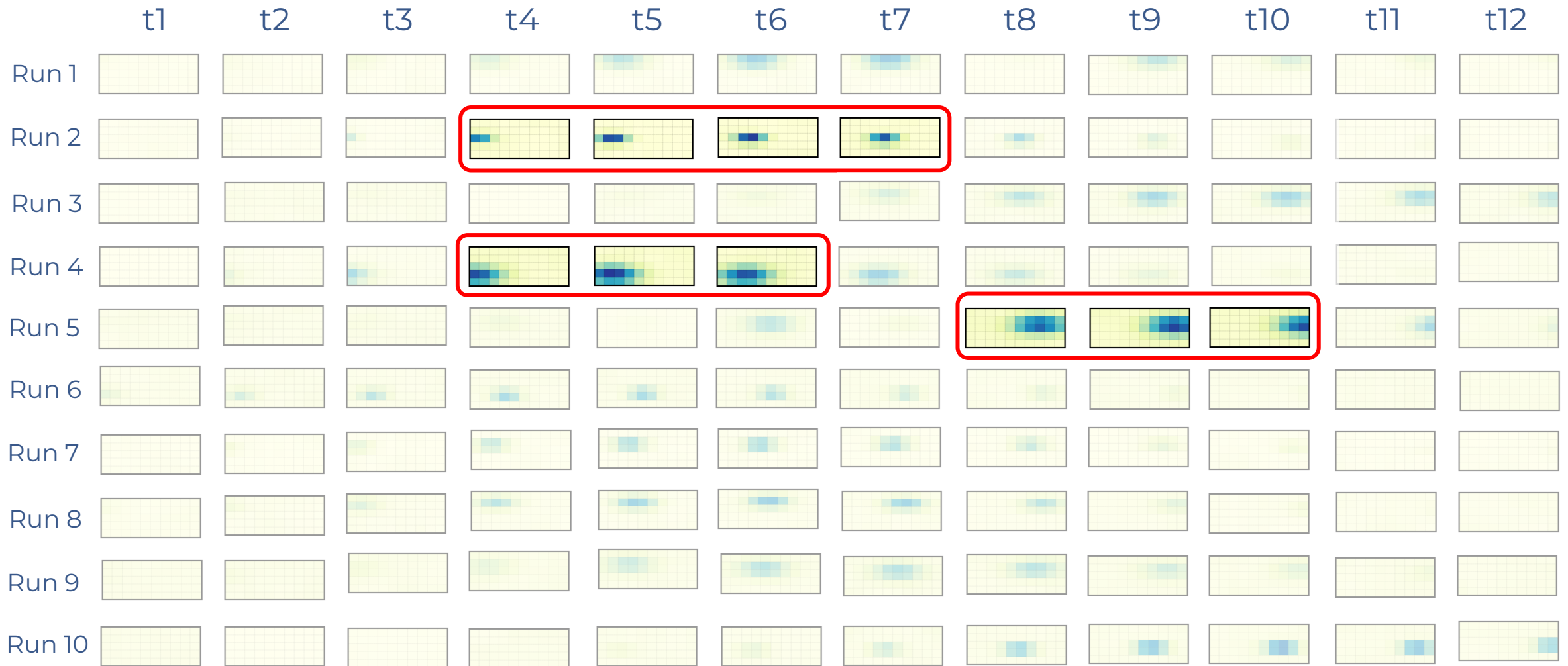


Run 1 — window=1h

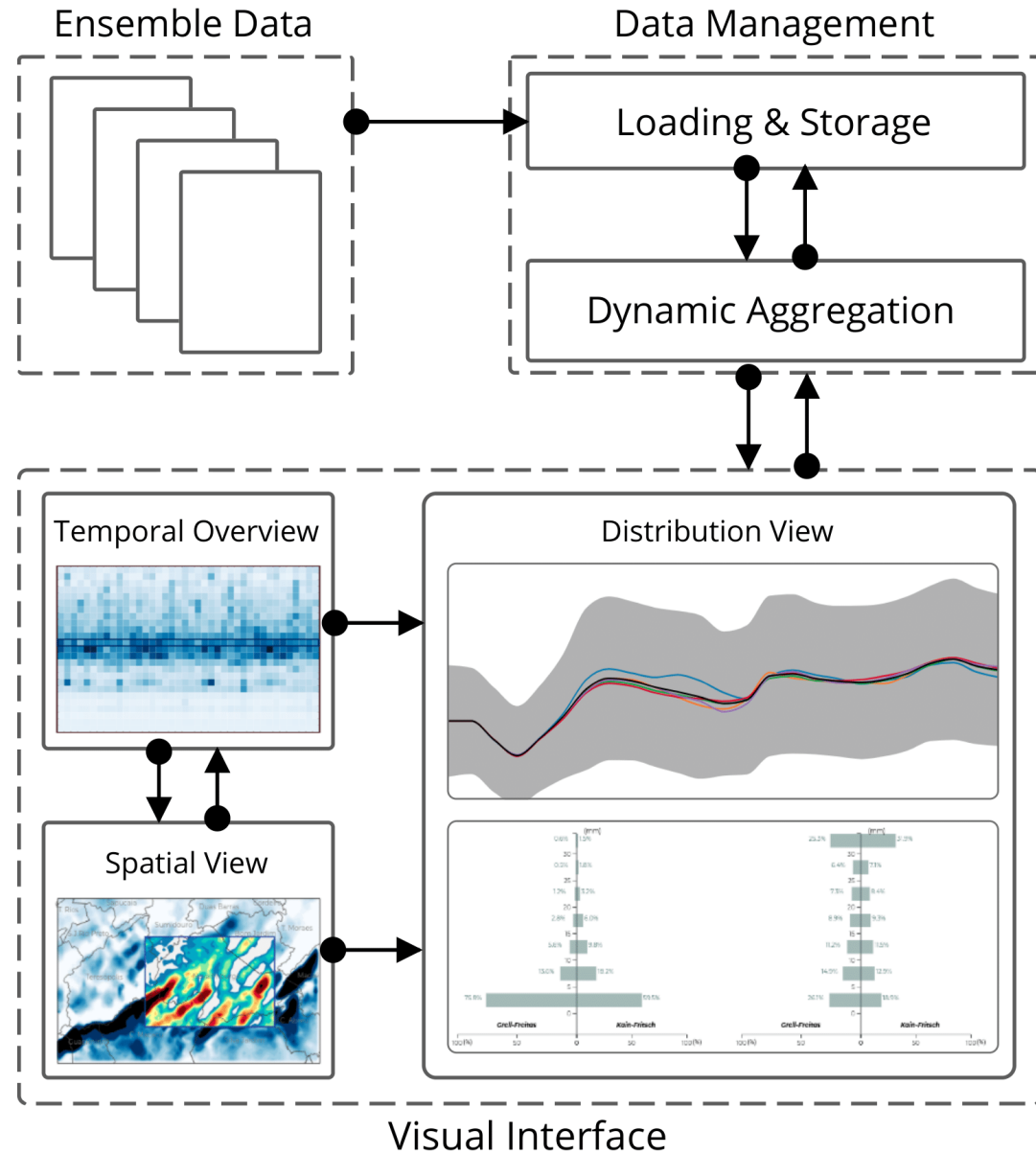


Accumulated rain — spatial max per sim/time (window=1)





X-Weather



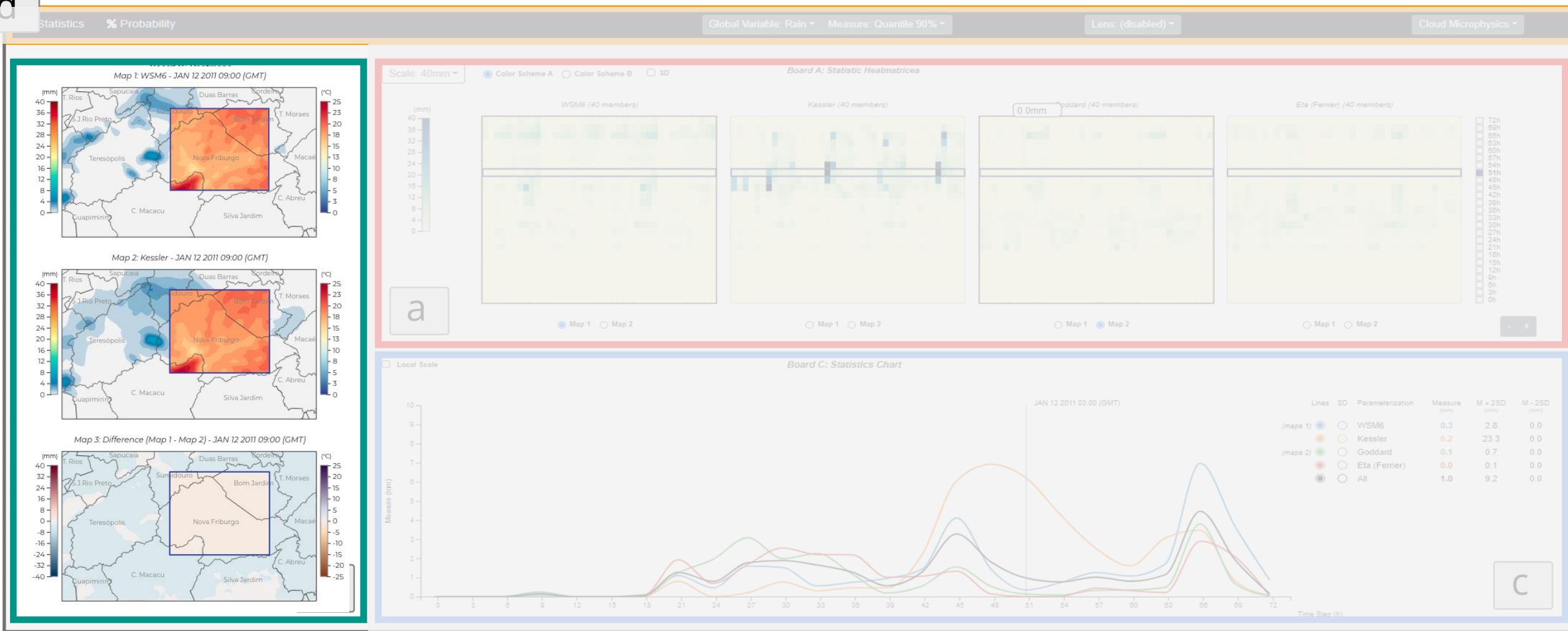
d



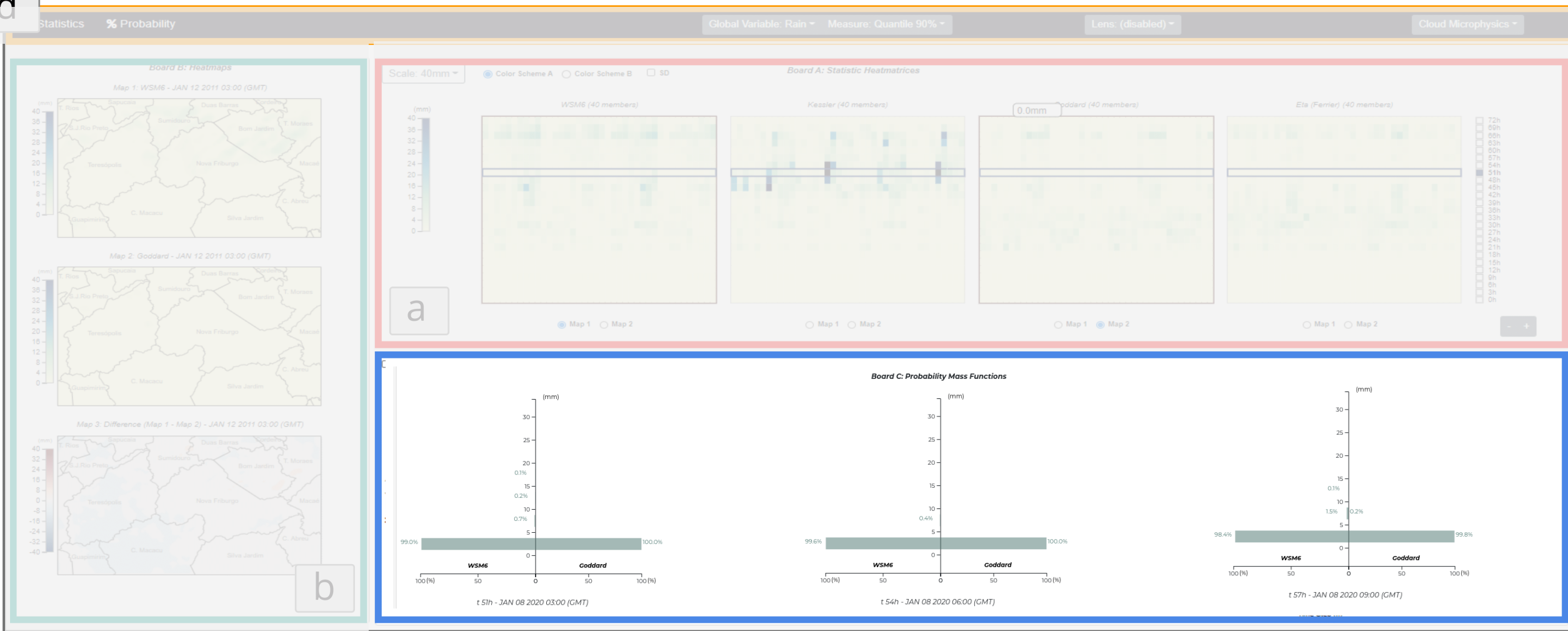
a



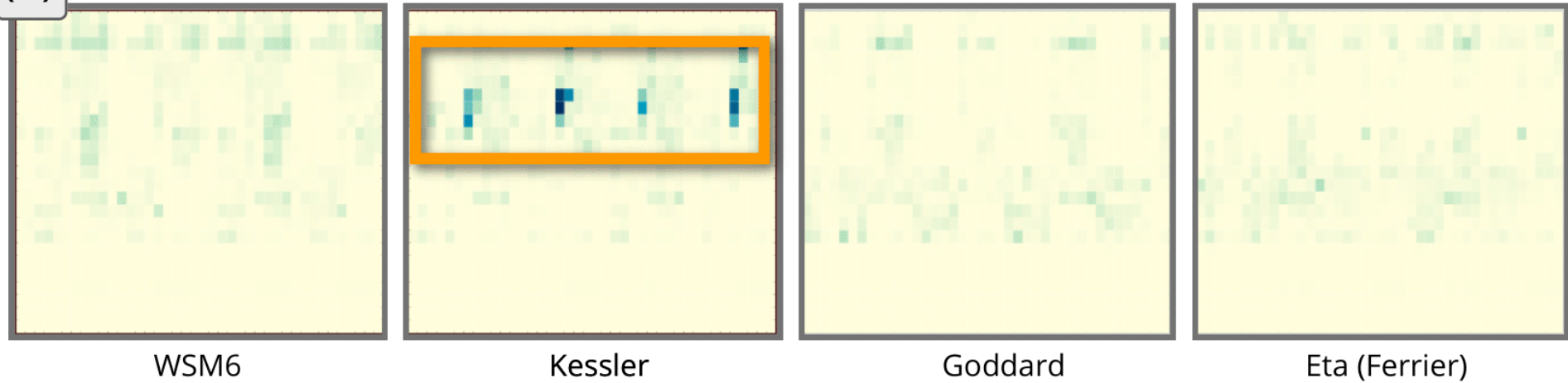
a



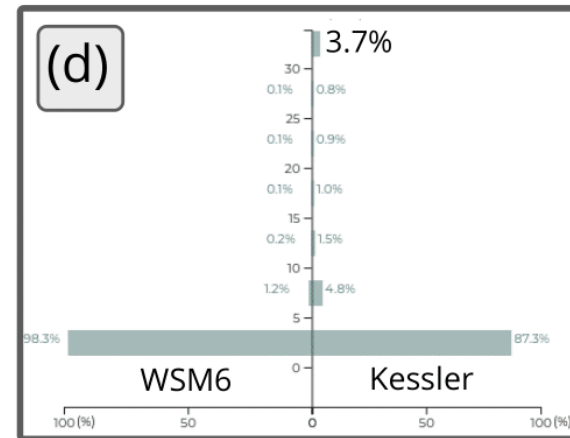
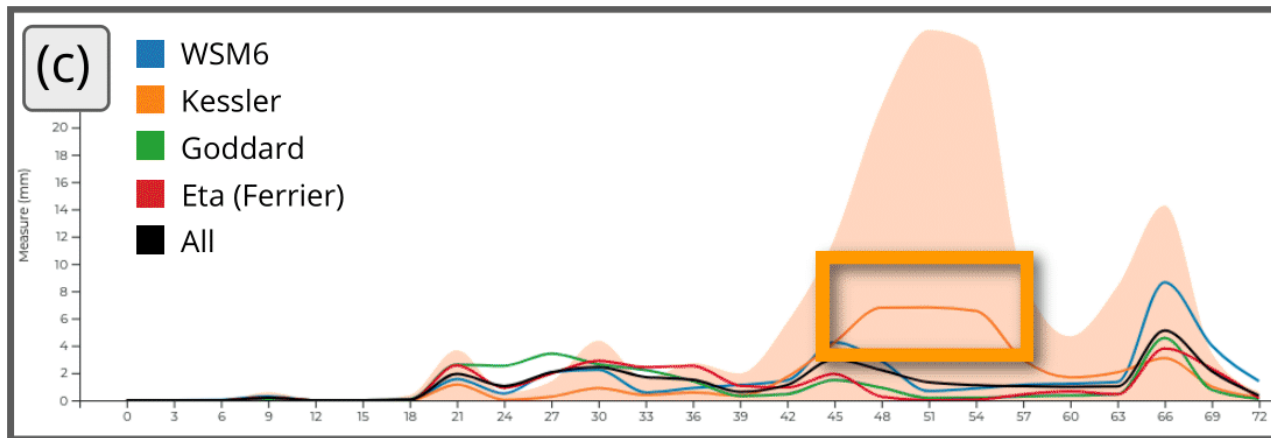
a



(b) Temporal View (Cloud Microphysics)

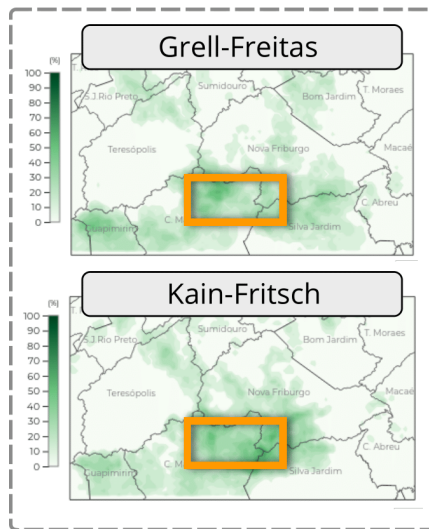


Distribution View (Cloud Microphysics)

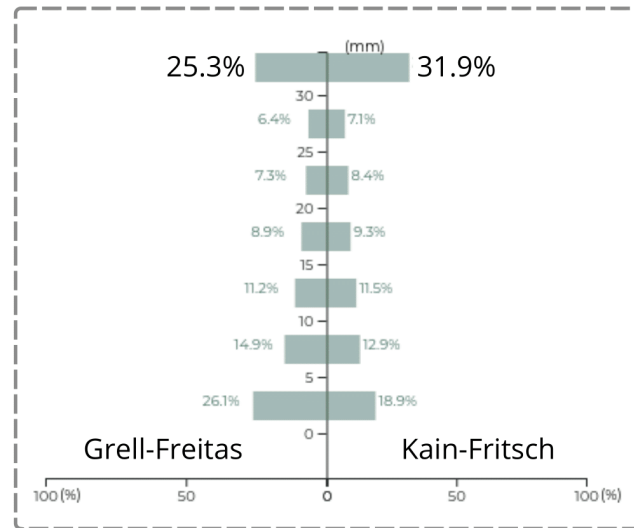


Accumulated Rain

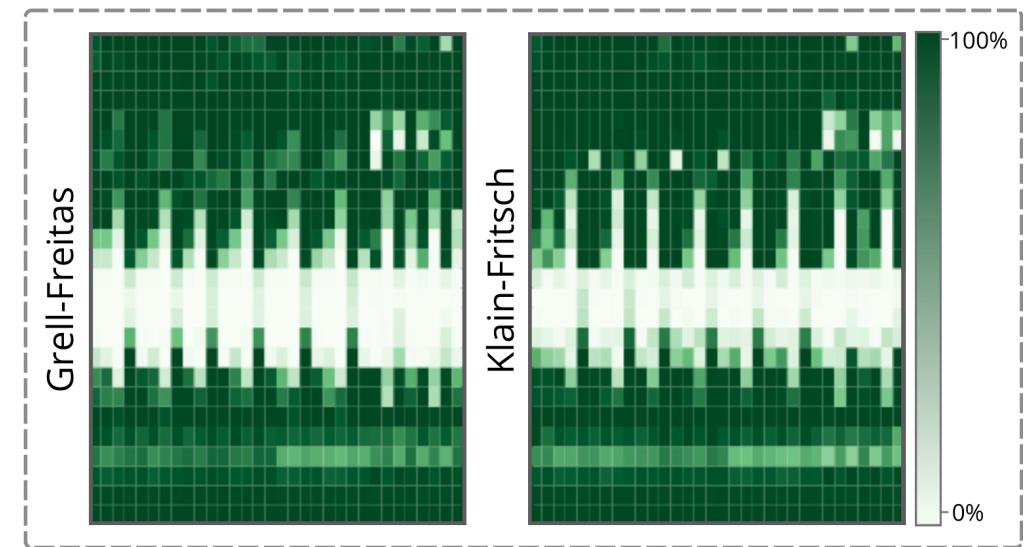
(a) $P(\text{Rain} \geq 30\text{mm})$



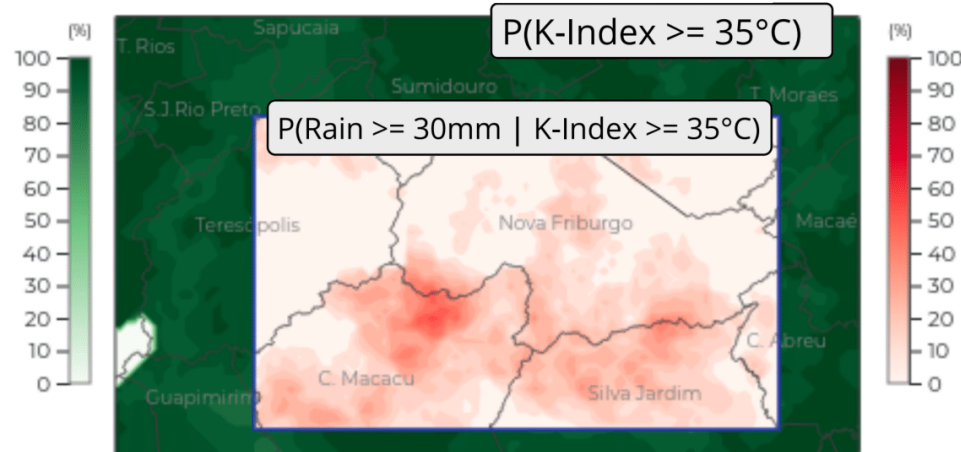
(b) Accumulated Rain Probability



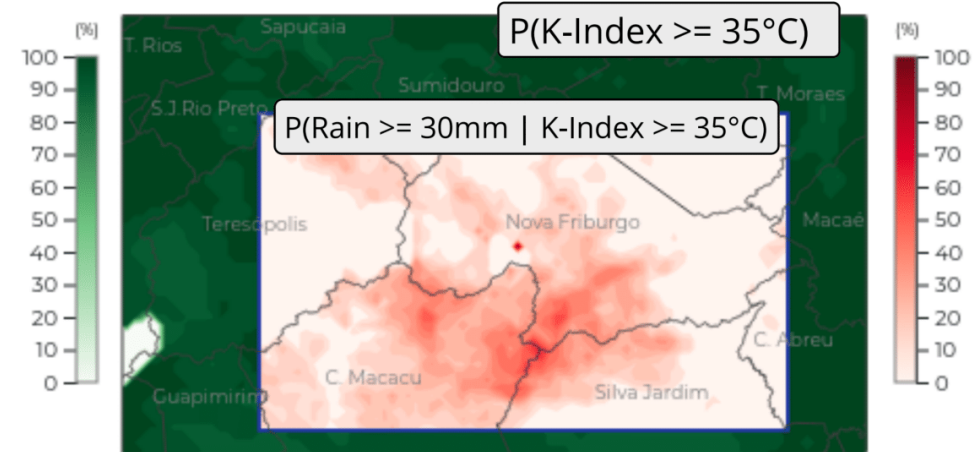
(c) $P(\text{K-Index} \geq 35^\circ\text{C})$



Grell-Freitas

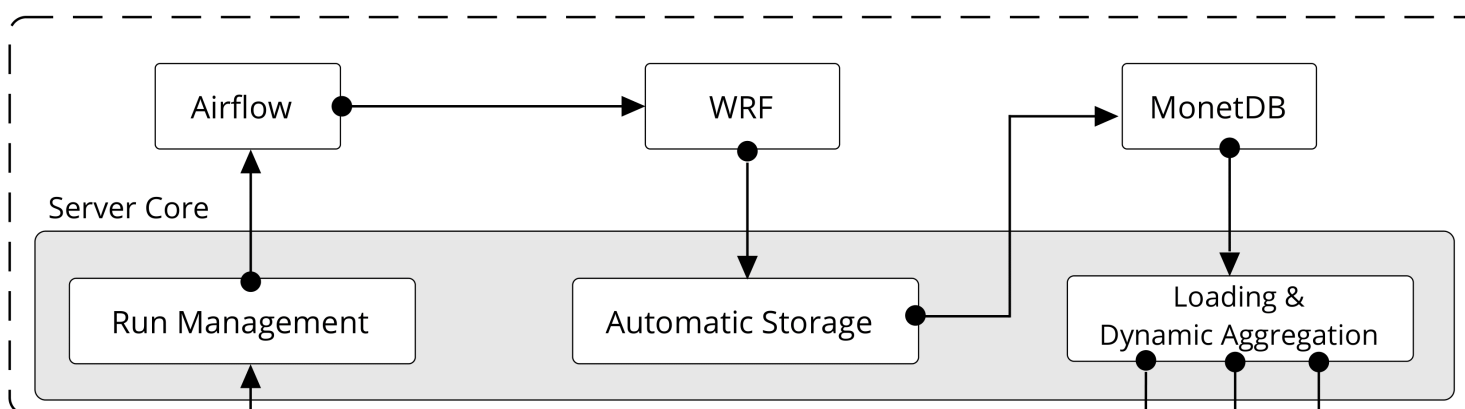


Kain-Fritsch



ProWis

Backend



Interface

